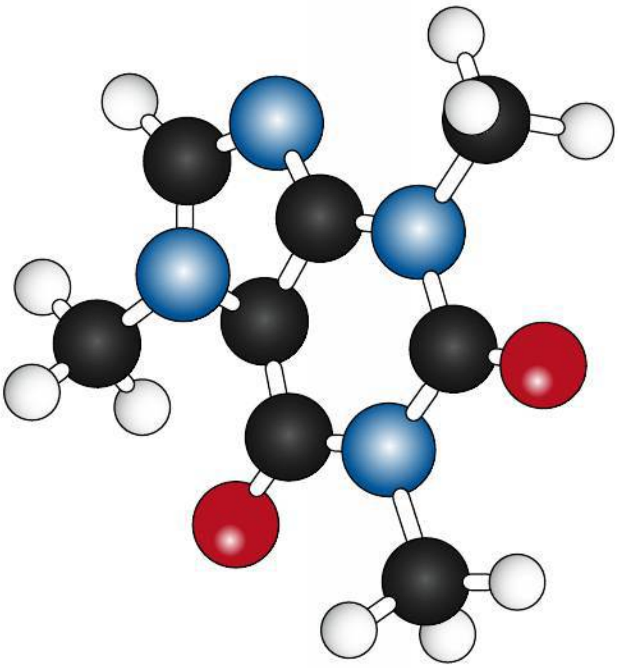


Figure 1.10 A molecular model of the stimulant, caffeine. The atoms are colour coded as follows: black = carbon, grey = hydrogen, red = oxygen, blue = nitrogen. Can you deduce the molecular formula, the empirical formula, and the molar mass of caffeine?

Solution

First calculate the mass of the empirical formula:

$$\text{mass}(HgCl) = 200.59 + 35.45 = 236.04\ \text{g mol}^{-1}$$

$$(236.04) \times x = M = 472.08$$

$$\therefore x = 2$$

$$\text{molecular formula} = Hg_2Cl_2$$

Combustion analysis usually gives data on the mass of compounds formed

The data presented so far may suggest that combustion analysis directly gives information on the relative masses of individual elements in a compound. In fact this is rarely the case, but instead elements are converted into new compounds, typically their oxides, by reaction with oxygen. So the primary data obtained are the masses of carbon dioxide, water, sulfur dioxide, and so on, which are measured by infra-red absorption, as described in Chapter 11. Processing these data simply involves an extra step.

Worked example

A 0.5438 g sample of a compound known to contain only carbon, hydrogen, and oxygen was burned completely in oxygen. The products were 1.0390 g CO_2 and 0.6369 g H_2O. Determine the empirical formula of the compound.

Solution

First we must convert the mass of each compound to moles in the usual way. From the number of moles of CO_2 and H_2O we can deduce the number of moles of C atoms and H atoms.

$$n(CO_2) = \frac{1.0390\ \text{g}}{12.01 + (16.00 \times 2)\ \text{g mol}^{-1}} = 0.02361\ \text{mol } CO_2 \Rightarrow 0.02361\ \text{mol C atoms}$$

$$n(H_2O) = \frac{0.6369\ \text{g}}{(1.01 \times 2) + 16.00\ \text{g mol}^{-1}} = 0.03534\ \text{mol } H_2O \Rightarrow 0.03534 \times 2 = 0.07068\ \text{mol H atoms}$$

In order to know the mass of O in the original sample, we must convert the number of moles of C and H atoms to mass by multiplying by their molar mass, *M*.

$$\text{mass C} = 0.02361\ \text{mol} \times 12.01\ \text{g mol}^{-1} = 0.2836\ \text{g}$$

$$\text{mass H} = 0.07068\ \text{mol} \times 1.01\ \text{g mol}^{-1} = 0.07139\ \text{g}$$

$$\therefore \text{mass O} = 0.5438 - (0.2836 + 0.07139) = 0.1888\ \text{g}$$

$$\text{mol O atoms} = \frac{0.1888\ \text{g}}{16.00\ \text{g mol}^{-1}} = 0.01180\ \text{mol}$$

Now we can proceed as with the previous examples, converting mass of O to moles and then comparing the mole ratios.

	Carbon, C	Hydrogen, H	Oxygen, O
mass / g			0.1888
moles	0.02361	0.07068	0.01180
divide by smallest	2.001	5.999	1.000
nearest whole number ratio	2	6	1

So the empirical formula is C_2H_6O.

Exercises

24 Give the empirical formulas of the following compounds:

- **(a)** ethyne, C_2H_2
- **(b)** glucose, $C_6H_{12}O_6$
- **(c)** sucrose, $C_{12}H_{22}O_{11}$
- **(d)** octane, C_8H_{18}
- **(e)** oct-1-yne, C_8H_{14}
- **(f)** ethanoic acid, CH_3COOH

25 A sample of a compound contains only the elements sodium, sulfur, and oxygen. It is found by analysis to contain 0.979 g Na, 1.365 g S, and 1.021 g O. Determine its empirical formula.

26 A sample of a hydrated compound was analysed and found to contain 2.10 g Co, 1.14 g S, 2.28 g O, and 4.50 g H_2O. Determine its empirical formula.

27 A street drug has the following composition: 83.89% C, 10.35% H, 5.76% N. Determine its empirical formula.

28 The following compounds are used in the production of fertilizers. Determine which has the highest percentage by mass of nitrogen: NH_3, $CO(NH_2)_2$, $(NH_4)_2SO_4$.

29 A compound has a formula M_3N where M is a metal element and N is nitrogen. It contains 0.673 g of N per gram of the metal M. Determine the relative atomic mass of M and so its identity.

30 Compounds of cadmium are used in the construction of photocells. Deduce which of the following has the highest percentage by mass of cadmium: CdS, CdSe, CdTe.

31 Benzene is a hydrocarbon, a compound of carbon and hydrogen only. It is found to contain 7.74% H by mass. Its molar mass is 78.10 g mol^{-1}. Determine its empirical and molecular formulas.

32 A weak acid has a molar mass of 162 g mol^{-1}. Analysis of a 0.8821 g sample showed the composition by mass is 0.0220 g H, 0.3374 g P, and the remainder was O. Determine its empirical and molecular formulas.

33 ATP is an important molecule in living cells. A sample with a mass of 0.8138 g was analysed and found to contain 0.1927 g C, 0.02590 g H, 0.1124 g N, and 0.1491 g P. The remainder was O. Determine the empirical formula of ATP. Its formula mass was found to be 507 g mol^{-1}. Determine its molecular formula.

34 A 0.30 g sample of a compound that contains only carbon, hydrogen, and oxygen was burned in excess oxygen. The products were 0.66 g of carbon dioxide and 0.36 g of water. Determine the empirical formula of the compound.

35 You are asked to write your name on a suitable surface, using a piece of chalk that is pure calcium carbonate, $CaCO_3$. How could you calculate the number of carbon atoms in your signature?

How many atoms of calcium are in Avogadro's name written in chalk?

1.3 Reacting masses and volumes

Understandings:

- Reactants can be either limiting or excess.
- The experimental yield can be different from the theoretical yield.
- Avogadro's law enables the mole ratio of reacting gases to be determined from volumes of the gases.
- The molar volume of an ideal gas is a constant at specified temperature and pressure.

  Guidance
  Values for the molar volume of an ideal gas are given in the IB data booklet in Section 2.
- The molar concentration of a solution is determined by the amount of solute and the volume of solution.

  Guidance
  The use of square brackets to denote molar concentration is required.
- A standard solution is one of known concentration.

Applications and skills:

- Solution of problems relating to reacting quantities, limiting and excess reactants, and theoretical, experimental, and percentage yields.
- Calculation of reacting volumes of gases using Avogadro's law.
- Solution of problems and analysis of graphs involving the relationship between temperature, pressure, and volume for a fixed mass of an ideal gas.
- Solution of problems relating to the ideal gas equation.

  Guidance
  The ideal gas equation, $PV = nRT$, and the value of the gas constant (R), are given in the IB data booklet in Sections 1 and 2.
- Explanation of the deviation of real gases from ideal behaviour at low temperature and high pressure.
- Obtaining and using experimental values to calculate the molar mass of a gas from the ideal gas equation.
- Solution of problems involving molar concentration, amount of solute, and volume of solution.

  Guidance
  Units of concentration to include: g dm^{-3}, mol dm^{-3}, and parts per million (ppm).
- Use of the experimental method of titration to calculate the concentration of a solution by reference to a standard solution.

Chemical equations show reactants combining in a fixed molar ratio

Chemical change, as summarized in chemical equations, is simply an expression of reactants combining in fixed ratios to form products. The most convenient means to express this ratio is as moles, as that gives us a means of relating the number of particles that react to the mass that we can measure. So, for example, when methane, CH_4, burns in air, we can conclude the following, all from the balanced chemical equation:

	$CH_4(g)$	+ $2O_2(g)$	→	$CO_2(g)$	+ $2H_2O(g)$
Reacting ratio by mole:	1 mole	2 moles		1 mole	2 moles
Reacting ratio by mass:	16.05 g	64.00 g		44.01 g	36.04 g
	80.05 g reactant			80.05 g product	

(The figures for total mass of reactant and product are just a check, as we know something would be wrong if they did not equate.)

The Bunsen burner controls the combustion of methane, CH_4, and is often used as a source of heat in the laboratory.

This simple interpretation of equations, going directly from coefficients to molar ratios, opens the door to a wide range of calculations involving reacting masses.

Worked example

Calculate the mass of carbon dioxide produced from the complete combustion of 1.00 g of methane.

Solution

Write the balanced equation and deduce the mole ratio as above. Then pick out from the question the terms that we need to analyse, here they are marked in red; these are the species where we need to convert moles to grams.

	$CH_4(g)$	+	$2O_2(g)$	$\rightarrow$	$CO_2(g)$	+	$2H_2O(g)$
Reacting ratio by mole:	1 mole				1 mole		
Reacting ratio by mass / g:	16.05				44.01		
For 1.00 g methane:	1.00				x		

Now solve the ratio, shown here using cross-multiplication, to determine the value of x.

$$\frac{\text{g } CH_4}{\text{g } CO_2} = \frac{16.05}{44.01} = \frac{1.00}{x}$$

$$x = \frac{1.00 \times 44.01}{16.05} = 2.74 \text{ g } CO_2$$

All questions on reacting ratios involve a variation of this approach:

- write the balanced equation;
- work out the mole ratio for the species identified in the question;
- work out the reacting ratio by mass for these species, using $m = n$ M;
- insert the data from the question and solve the ratio.

Worked example

Iodine chloride, ICl, can be made by the following reaction:

$$2I_2 + KIO_3 + 6HCl \rightarrow 5ICl + KCl + 3H_2O$$

Calculate the mass of iodine, I_2, needed to prepare 28.60 g of ICl by this reaction.

Solution

The relevant terms from the question are I_2 and ICl, so these are our focus.

	$2I_2$	+	KIO_3	+	$6HCl$	$\rightarrow$	$5ICl$	+	KCl	+	$3H_2O$
reacting ratio by mole:	2 moles						5 moles				
reacting ratio by mass:	2 × (126.90 × 2) = 507.60 g						5 × (126.90 + 35.45) = 811.75 g				
For 28.60 g ICl:	x						28.60 g				

$$\frac{\text{g } I_2}{\text{g ICl}} = \frac{507.60}{811.75} = \frac{x}{28.60}$$

$$x = \frac{507.60 \times 28.60}{811.75} \text{ g } I_2 = 17.88 \text{ g } I_2$$

The term **carbon footprint** refers to the mass of carbon dioxide and other greenhouse gases such as methane that an individual emits in a 1-year period. It is often expressed as the carbon dioxide equivalent, or CO_2e, to represent the total climate change impact of all the greenhouse gases caused by an item or activity. It includes emissions from fuels used in transport, services such as heating, production and consumption of food, and the direct and indirect emissions from manufactured goods and construction. It is extremely difficult to measure all sources accurately, but the fundamental concept uses the type of calculation shown here. The carbon footprint is a measure of an individual's consumption of resources, and suggests the link between this and the enhanced greenhouse effect. There is further discussion of carbon footprints in Chapter 12.

Remember to convert between mass and moles by using $n = \frac{m}{M}$ or $m = n\,M$

It is common for people to waste time solving questions like this by doing the mole to gram conversions for *all* the species represented in the equation. You can save yourself a lot of trouble by focusing *only* on the terms indicated in the question, as shown in this example.

The theoretical yield is determined by the limiting reactant

Imagine that you are following a recipe to make 12 cookies. It calls for you to mix two eggs with four cups of flour. The problem is that you only have one egg. You will quickly realize that this means you can use only two cups of flour and end up with only six cookies. We could say that the number of eggs *limited* the amount of product.

In many chemical reactions the relative amounts of reactants available to react together will similarly affect the amount of product. The reactant that determines the quantity of product is known as the **limiting reactant**. Other reactants will therefore not be fully used, and are said to be in **excess**. Identifying the limiting reactant is therefore a crucial step before we can calculate the expected quantity of product. The **theoretical yield**, which is usually expressed in grams or moles, refers to the maximum amount of product obtainable, assuming 100% of the limiting reactant is converted to product.

Note that identification of the limiting reactant depends on the *mole ratios* in the balanced chemical equation for the reaction. This means that if reactant quantities are given in grams, they must first be converted to moles.

The concept of limiting reactant is often useful in the design of experiments and synthetic processes. By deliberately making one reactant available in an amount greater than that determined by its mole ratio in the balanced equation, it ensures that the other reactant is limiting and so will be fully used up. For example, in order to remove lead ions in lead nitrate from a contaminated water supply, sodium carbonate, Na_2CO_3, is added to precipitate lead carbonate, as shown in the equation:

$$\underbrace{Pb(NO_3)_2(aq)}_{\text{limiting}} + \underbrace{Na_2CO_3(aq)}_{\text{excess}} \rightarrow PbCO_3(s) + 2NaNO_3(aq)$$

By using excess Na_2CO_3, this ensures that *all* the lead ions react and so are removed from the water supply.

Worked example

Nitrogen gas (N_2) can be prepared from this reaction:

$$2NH_3(g) + 3CuO(s) \rightarrow N_2(g) + 3Cu(s) + 3H_2O(g)$$

If 18.1 g NH_3 are reacted with 90.40 g CuO, determine the mass of N_2 that can be formed.

Solution

First we must determine the limiting reactant. We convert the mass of reactants to moles, and then compare the mole ratio in the balanced equation with the mole ratio of reactants given.

$$n(NH_3) = \frac{18.1\ \text{g}}{14.01 + (3 \times 1.01)\ \text{g mol}^{-1}} = 1.06\ \text{mol}\ NH_3$$

$$n(CuO) = \frac{90.40\ \text{g}}{63.55 \times 16.00\ \text{g mol}^{-1}} = 1.14\ \text{mol CuO}$$

$$\text{mole ratio from equation: } \frac{NH_3}{CuO} = \frac{2}{3} = 0.667$$

$$\text{mole ratio from given masses: } \frac{NH_3}{CuO} = \frac{1.06}{1.14} = 0.930$$

As the ratio NH_3 : CuO of the given masses is *larger* than the required ratio in the equation, it means NH_3 is in excess and CuO is the limiting reactant.

This means that the amount of N_2 that can form will be determined by the amount of CuO. This is now similar to the earlier questions, where we write out the equation and focus on the terms identified in the question.

	$2NH_3(g)$ +	$3CuO(s)$ →	$N_2(g)$ +	$3Cu(s)$ + $3H_2O(g)$
reacting ratio by mole:		3 moles	1 mole	
for 1.14 moles CuO:		1.14	x	

$$\text{mole ratio } \frac{CuO}{N_2} = \frac{3}{1} = \frac{1.14}{x} \qquad \therefore x = \frac{1.14 \times 1}{3} = 0.380\ \text{mol}\ N_2$$

$$\therefore \text{mass } N_2 = 0.380\ \text{mol} \times M(N_2)\ \text{g mol}^{-1} = 0.380\ \text{mol} \times 28.02\ \text{g mol}^{-1} = 10.7\ \text{g}\ N_2$$

There are alternate approaches to determining the limiting reactant, such as calculating which given amount of reactant would yield the smallest amount of product. But in essence, all questions on limiting reactant and theoretical yield involve comparing the mole ratio of given masses of reactants with the coefficients in the equation. This is a summary of the steps:

- write the balanced equation and focus on the mole ratio of reactants;
- convert the given mass of reactants to moles;
- compare the given mole ratios with the ratio of coefficients in the equation;
- identify the limiting reactant from the above ratios;
- calculate the moles of product from the given moles of limiting reactant.

The limiting reactant determines the amount of product that can form. The theoretical yield is the quantity of product that can form from the complete conversion of the limiting reactant.

Sometimes it is useful to measure how much excess reactant will remain when all the limiting reactant has been used up and the reaction stops. One example of this is a technique called **back-titration**, which analyses excess acid or alkali after a reaction is complete, and so indirectly measures the amount of a limiting reactant. This is explained on page 50.

A simple example of how to calculate the excess is shown below, using an example of burning CH_4 when 1 mole of CH_4 and 1 mole of O_2 are supplied.

	$CH_4(g)$ +	$2O_2(g)$ →	$CO_2(g)$ +	$2H_2O(g)$
reacting ratio by mole:	1	2		
reactant ratio given:	1	1		
⇒ O_2 is limiting reactant				
mole ratio at the end of reaction:	0.5	0	0.5	1

So 0.5 moles CH_4 will be un-reacted at the end of the reaction.

We will study more examples of this type of question in the next section.

The percentage yield can be calculated from the experimental and theoretical yields

The answer to the Worked example above, 10.7 g N_2 is known as the theoretical yield, because it assumes that *all* of the CuO was converted to N_2 with no loss, impurities present, wastage, or incomplete reaction. In reality, all of the above happen to different extents in most chemical reactions, and so the theoretical yield is usually different from the actual or **experimental yield**.

When we compare the experimental yield with the theoretical yield, we get a measure of the efficiency of the conversion of reactants to products. This is usually expressed as the **percentage yield**, defined as follows:

$$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$$

In your own experiments, you may often be able to calculate the percentage yield of product in evaluating the results. In industry, this is a very important calculation to determine the efficiency of a process such as the synthesis of a drug in the pharmaceutical industry. Many aspects of green chemistry focus on ways to increase the yield of product by reducing wastage.

▲ Industrial plants such as this oil refinery need to be able to track the efficiency of the chemical reactions taking place. Measuring the yield of product is an essential part of this.

CHALLENGE YOURSELF

4 Percentage yield and the atom economy are different concepts, but both can be used to assess aspects of the overall efficiency of a chemical process. See if you can find a reaction that has a high percentage yield under certain conditions, but a low atom economy.

Worked example

The previous Worked example on the synthesis of N_2 from NH_3 and CuO had a theoretical yield of 10.7 g N_2 from the starting amounts of reactants. Under the same conditions, an experiment produced 8.35 g N_2. Determine the percentage yield.

Solution

$$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$$

$$\therefore \text{percentage yield} = \frac{8.35\text{ g}}{10.7\text{ g}} \times 100 = 78.0\%$$

Exercises

36 Iron ore can be reduced to iron by the following reaction:

$$Fe_2O_3(s) + 3H_2(g) \rightarrow 2Fe + 3H_2O(l)$$

(a) How many moles of Fe can be made from 1.25 moles of Fe_2O_3?
(b) How many moles of H_2 are needed to make 3.75 moles of Fe?
(c) If the reaction yields 12.50 moles of H_2O, what mass of Fe_2O_3 was used up?

37 Lighters commonly use butane, C_4H_{10}, as the fuel.

(a) Formulate the equation for the combustion of butane.
(b) Determine the mass of butane that burned when 2.46 g water were produced.

38 Booster rockets for the space shuttle use the following reaction:

$$3Al(s) + 3NH_4ClO_4(s) \rightarrow Al_2O_3(s) + AlCl_3(s) + 3NO(g) + 6H_2O(g)$$

Calculate the mass of NH_4ClO_4 that should be added to this fuel mixture to react completely with every kilogram of Al.

39 Limestone is mostly calcium carbonate, $CaCO_3$, but also contains other minerals. When heated, the $CaCO_3$ decomposes into CaO and CO_2. A 1.605 g sample of limestone was heated and gave off 0.657 g of CO_2.

(a) Formulate the equation for the thermal decomposition of calcium carbonate.
(b) Determine the percentage mass of $CaCO_3$ in the limestone.
(c) State the assumptions that you are making in this calculation.

40 Methanol, CH_3OH, is a useful fuel that can be made as follows:

$$CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$$

A reaction mixture used 12.0 g of H_2 and 74.5 g of CO.

(a) Determine the theoretical yield of CH_3OH.
(b) Calculate the amount of the excess reactant that remains unchanged at the end of the reaction.

41 The dry-cleaning solvent 1,2-dichloroethane, $C_2H_4Cl_2$, is prepared from the following reaction:

$$C_2H_4(g) + Cl_2(g) \rightarrow C_2H_4Cl_2(l)$$

Determine the mass of product that can be formed from 15.40 g of C_2H_4 and 3.74 g of Cl_2.

42 Calcium carbonate, $CaCO_3$, is able to remove sulfur dioxide, SO_2, from waste gases by a reaction in which they react in a 1: 1 stoichiometric ratio to form equimolar amounts of $CaSO_3$. When 255 g of $CaCO_3$ reacted with 135 g of SO_2, 198 g of $CaSO_3$ were formed. Determine the percentage yield of $CaSO_3$.

43 Pentyl ethanoate, $CH_3COOC_5H_{11}$, which smells like bananas, is produced from the esterification reaction:

$$CH_3COOH(aq) + C_5H_{11}OH(aq) \rightarrow CH_3COOC_5H_{11}(aq) + H_2O(l)$$

A reaction uses 3.58 g of CH_3COOH and 4.75 g of $C_5H_{11}OH$ and has a yield of 45.00%. Determine the mass of ester that forms.

44 A chemist has to make a 100 g sample of chlorobenzene, C_6H_5Cl, from the following reaction:

$$C_6H_6 + Cl_2 \rightarrow C_6H_5Cl + HCl$$

Determine the minimum quantity of benzene, C_6H_6, that can be used to achieve this with a yield of 65%.

Avogadro's law directly relates gas volumes to moles

All the examples above use mass as a way to measure amount, the number of moles. But in the laboratory we often work with liquids and gases, where *volume* is a more convenient measure. So what is the relationship between gas volume and number of moles?

Consider the following demonstration, where two gas jars are each filled with different gases – hydrogen (H_2) in flask A and bromine (Br_2) in flask B. The flasks are at the same temperature and pressure and have equal volumes.

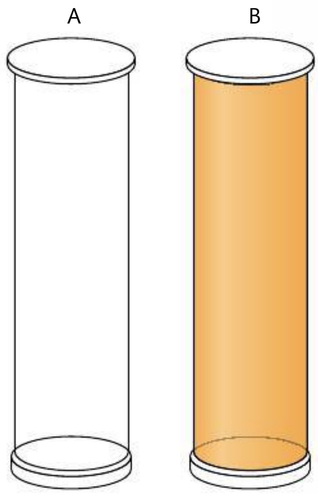


Figure 1.11 Flask A contains hydrogen molecules, flask B contains bromine molecules. The two flasks are under the same conditions of temperature and pressure.

It is known, from many experimental measurements on gas volumes, that the number of particles in the two flasks above is the same. At first this might seem surprising – after all bromine molecules are much larger and heavier than hydrogen molecules. But we need to consider the nature of the gaseous state, and as we learned on page 9, remember that the particles in a gas are widely spaced out with only negligible forces between them. In simple terms, most of a gas volume is empty space. And for this reason the chemical nature of the gas is irrelevant to its volume. Gas volume is determined only by the number of particles and by the temperature and pressure.

This understanding is known as Avogadro's law, which states that:

Equal volumes of all gases, when measured at the same temperature and pressure, contain an equal number of particles.

Alternatively, it can be stated that equal numbers of particles of all gases, when measured at the same temperature and pressure, occupy equal volumes.

Using V for volume and n for number of moles:

$$V \propto n$$

This relationship enables us to relate gas volumes (of any gas) to the number of moles, and so to reacting ratios in equations.

Worked example

40 cm^3 of carbon monoxide are reacted with 40 cm^3 of oxygen in the reaction:

$$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$$

What volume of carbon dioxide is produced? (Assume all volumes are measured at the same temperature and pressure.)

Airbags have become a standard safety fitting in many vehicles. They are designed to act as a cushion or shock absorber by inflating rapidly on sudden impact of the vehicle during a collision. Airbags work on the principle of a chemical reaction triggered by the impact producing a gaseous product that causes a sudden volume change. The key reaction used is the conversion of sodium azide, NaN_3, to nitrogen gas, N_2. To avoid the production of dangerously reactive sodium metal, potassium nitrate, KNO_3, and silicon dioxide, SiO_2, are also included so that harmless silicates are produced instead.

Illustration of an airbag and seatbelt in action during a car accident. On impact, the airbag inflates and the seatbelt slows the forward force of the body, protecting the driver's head and chest.

Avogadro's law states that equal volumes of all gases at the same conditions of temperature and pressure contain equal numbers of particles: $V \propto n$.

CHALLENGE YOURSELF

5 Use the explanation on page 33 to deduce the chemical equations for the reactions taking place in a deployed airbag.

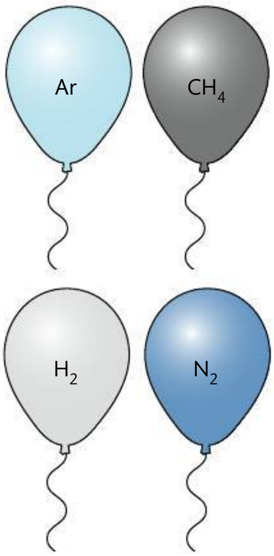


Figure 1.12 These four balloons are all filled with 1 dm^3 of gas. At 25 °C and 100 kPa, they each contain 0.044 mol or 2.65 × 10^{22} atoms or molecules. Which balloon is the heaviest?

Note that STP (standard temperature and pressure) is not the same as the 'standard state', which is used in thermodynamic data and is explained in Chapter 5.

STP refers to a temperature of 273 K and a pressure of 100 kPa.

Solution

First identify the mole ratios in the equation:

$$\begin{array}{ccccc} 2CO(g) & + & O_2(g) & \rightarrow & 2CO_2(g) \\ \text{2 moles} & & \text{1 mole} & & \text{2 moles} \end{array}$$

The mole ratio is equal to the ratio of reacting gas volumes, so:

$$\begin{array}{ccccc} 2CO(g) & + & O_2(g) & \rightarrow & 2CO_2(g) \\ 40\ \text{cm}^3 & & 20\ \text{cm}^3 & & 40\ \text{cm}^3 \end{array}$$

Therefore 40 cm^3 of carbon dioxide are produced. (Oxygen is in excess by 20 cm^3.)

Worked example

When 10 cm^3 of a gaseous hydrocarbon (a compound containing only carbon and hydrogen) is burned in excess oxygen, the products consist of 30 cm^3 of carbon dioxide and 30 cm^3 of water vapour, measured under the same conditions of temperature and pressure. Determine the molecular formula of the hydrocarbon.

Solution

'Excess' oxygen indicates that the combustion reaction is complete.

$$\begin{array}{lccccccc} & C_xH_y & + & \text{excess } O_2 & \rightarrow & CO_2 & + & H_2O \\ \text{volumes:} & 10\ \text{cm}^3 & & & & 30\ \text{cm}^3 & & 30\ \text{cm}^3 \\ \text{ratio of volumes / mole ratio:} & 1 & & & & 3 & & 3 \end{array}$$

$$\therefore \text{1 molecule hydrocarbon} \rightarrow \underset{\text{3 C atoms}}{\text{3 molecules } CO_2} + \underset{\text{6 H atoms}}{\text{3 molecules } H_2O}$$

The molecular formula is C_3H_6.

All gases under the same conditions have the same molar volume

On the basis of Avogadro's law, the volume occupied by one mole of any gas, known as the **molar volume**, must be the same for *all* gases when measured under the same conditions of temperature and pressure.

At standard temperature and pressure (STP), one mole of a gas has a volume of 2.27 × 10^{-2} m^3 mol^{-1} (=22.7 dm^3 mol^{-1}). The conditions at STP are:

- a temperature of 0 °C (273 K)
- pressure of 100 kPa.

The molar volume can be used to calculate the amount of gas in a similar way to the use of molar mass earlier in this chapter. Here though the calculations are easier, as all gases have the same molar volume under the same conditions.

$$\text{number of moles of gas } (n) = \frac{\text{volume } (V)}{\text{molar volume}}$$

Worked example

What volume of oxygen at standard temperature and pressure would be needed to completely burn 1 mole of butane, C_4H_{10}?

Solution

As always, start with the balanced equation and pick out the terms from the question.

$$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(g)$$

mole / volume ratio: 1 6.5

6.5 moles of gas at STP have volume = 6.5 mol × 22.7 $dm^3\ mol^{-1}$ = 147.6 dm^3

Worked example

Calculate the volume occupied by 0.0200 g of He at standard temperature and pressure.

Solution

First convert the mass of He to moles.

$$n = \frac{m}{M} = \frac{0.0200\ \text{g}}{4.00\ \text{g mol}^{-1}} = 0.00500\ \text{mol}$$

$$\text{volume} = 0.00500\ \text{mol} \times 22.7\ \text{dm}^3\ \text{mol}^{-1} = 0.114\ \text{dm}^3$$

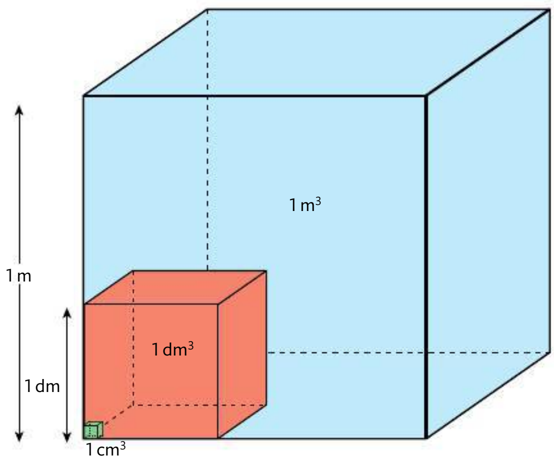
A note about units of volume

The metric unit m^3 is widely used in industrial and engineering calculations, but is too large to be convenient for many volume measurements in the laboratory. Instead, dm^3 and cm^3 are commonly used, so it is important to be able to interconvert these.

$$1\ \text{dm}^3 = 10^{-3}\ \text{m}^3 \qquad 1000\ \text{dm}^3 = 1\ \text{m}^3$$

$$1\ \text{cm}^3 = 10^{-3}\ \text{dm}^3 \qquad 1000\ \text{cm}^3 = 1\ \text{dm}^3$$

$$\text{cm}^3 \underset{\text{multiply by 1000}}{\overset{\text{divide by 1000}}{\rightleftarrows}} \text{dm}^3 \underset{\text{multiply by 1000}}{\overset{\text{divide by 1000}}{\rightleftarrows}} \text{m}^3$$



The litre (l or L) is widely used in place of dm^3, and millilitre (ml or mL) in place of cm^3. You will not be penalized for the use of these terms in an examination, but they will not be used in examination questions so it is essential that you know the correct use of m^3, dm^3, and cm^3.

The value of the molar volume and the conditions for STP are given in section 2 of the IB data booklet.

CODATA (Committee on Data for Science and Technology) is an interdisciplinary scientific committee of the International Council of Science. It was established in 1966 to promote the worldwide compilation and sharing of reliable numerical data, such as the molar volume of a gas.

Figure 1.13 The three cubes are not to scale. 1 m^3 = 1000 dm^3 = 1 000 000 cm^3. In the laboratory, volumes are usually measured in cm^3 or dm^3 and often these measurements need to be converted to m^3 in calculations.

Exercises

45 How many moles are present in each of the following at STP?

(a) 54.5 dm^3 CH_4 **(b)** 250.0 cm^3 CO **(c)** 1.0 m^3 O_2

46 What is the volume of each of the following at STP?

(a) 44.00 g N_2 **(b)** 0.25 mol NH_3

47 Pure oxygen gas was first prepared by heating mercury(II) oxide, HgO.

$$2HgO(s) \rightarrow 2Hg(l) + O_2(g)$$

What volume of oxygen at STP is released by heating 12.45 g of HgO?

48 Which sample contains more molecules, 3.14 dm^3 of bromine, Br_2, or 11.07 g of chlorine, Cl_2 when measured at the same temperature and pressure?

49 Calcium reacts with water to produce hydrogen.

$$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$$

Calculate the volume of gas at STP produced when 0.200 g of calcium reacts completely with water.

50 Dinitrogen oxide, N_2O, is a greenhouse gas produced from the decomposition of artificial nitrate fertilizers. Calculate the volume at STP of N_2O produced from 1.0 g of ammonium nitrate, when it reacts according to the equation:

$$NH_4NO_3(s) \rightarrow N_2O(g) + 2H_2O(l)$$

NATURE OF SCIENCE

Early ideas on gas behaviour were suggested from the postulates of the kinetic theory, but could not advance without scientific evidence. This was provided by experimental work, mainly that of Boyle and Mariotte, Charles, and Gay-Lussac who contributed quantitative data based on testable predictions of how gases would respond to changes in temperature, volume, and pressure. In a fairly classic example of scientific process, the data supported the theory, and the theory explained the data. As a result there was wide acceptance of what became known as 'the gas laws' by the 18th century.

It is interesting to consider on the other hand why Avogadro's hypothesis was not widely accepted initially. Experiments led him to suggest that equal volumes of all gases at the same temperature and pressure contain the same number of molecules, but data to confirm this was somewhat lacking. In addition, his ideas conflicted with Dalton's atomic theory, which suggested that particles in gases could be only single atoms, not molecules as Avogadro proposed. It took the logical argument of Cannizzaro nearly 50 years later to show that Avogadro's hypothesis could be explained, and moreover used as a means to determine molecular mass. Following this, the relationship between gas volume and number of molecules became widely accepted and known as Avogadro's law. History has shown that the acceptance of scientific ideas by the scientific community is sometimes influenced by the time and manner of their presentation, as well as by their power to explain existing ideas.

The gas laws describe pressure, volume, and temperature relationships for all gases

The kinetic theory of matter, summarized on page 9, describes gases as largely empty space containing free moving particles of negligible volume having no inter-particle forces. This is often referred to as the **ideal gas** model. In effect it is an approximation, as no gas fits this description exactly, but it is nonetheless a useful means for predicting and interpreting the physical properties of gases under typical conditions of temperature and pressure. Later in this section we will explore situations when gas behaviour deviates from this ideal model.

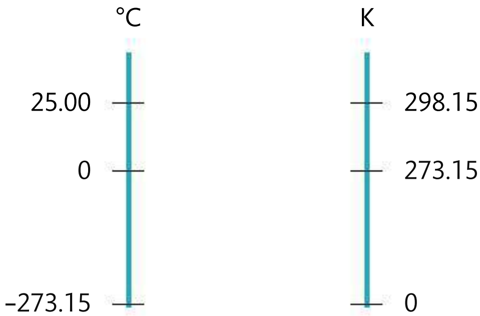
An ideal gas is one that obeys the ideal gas laws.

You will be familiar with some behaviour of gases through everyday experiences such as blowing up a balloon or inflating a bicycle tyre to increase the pressure. Perhaps you have noticed how inflated balloons shrivel in colder conditions and expand when it is warmer? The interesting thing about these simple observations of the volume, pressure, and temperature of a gas is that they are not dependent on the chemical nature of the gas. In fact, all gases respond in the same way to changes in volume, pressure, and temperature when the mass of gas is fixed. These relationships are summarized as the **gas laws** and are discussed below.

A note about units of temperature

In all work on gases, it is essential to use values for temperature recorded in Kelvin (K), not in Celsius (°C). Temperature in Kelvin is known as the **absolute temperature**, and is based on a scale where absolute zero, 0 K, is the point of zero kinetic energy of particles. This coincides with –273.15 °C. As the interval on the Kelvin scale is the same as that on the Celsius scale, conversion between the two simply involves addition or subtraction of 273.15 (commonly approximated to 273).

$$\text{temperature (K)} = \text{temperature (°C)} + 273.15$$



William Thomson (1824–1907), who became known as Lord Kelvin later in life, completed most of his work at the University of Glasgow, Scotland. His concept of the absolute temperature scale followed from his recognition of the relationship between heat energy and the ability to do work. The existence of a minimum possible temperature at which no heat can be extracted from the system and so no work done, led him to the definition of absolute zero in 1848. This in turn led to the formulation of the laws of thermodynamics. Kelvin is considered one of the great scientists of the 19th century, and is buried next to Isaac Newton in London.

A note about units of pressure

The SI unit of pressure is the Pascal (Pa), which is equal to N m^{-2}. Pressure is now commonly given in bars, where 10^5 Pa = 1 bar, as this is conveniently close to 1 atmospheric pressure.

NATURE OF SCIENCE

The definition of absolute zero (0 K) as the temperature where a substance has no kinetic energy, suggests it is the point at which all motion in particles ceases. It is the lowest possible temperature. Zero Kelvin has not been achieved, although modern technologies, which improve cooling methods and use magnets to contain the gas, are helping scientists to reach values ever closer to this. Researchers in Finland have achieved temperatures as low as 100 pK (1×10^{-10} K) in a piece of rhodium metal. Ultra-low temperature research has led to observations of phenomena such as quantum fluid behaviour and superconductivity, and could lead to improvements in precision measurements such as those used in atomic clocks and sensors for gravity and rotation. Science progresses as improvements in technology give access to new information, and studies in one field open up possibilities in another. Superconductivity is discussed further in Chapter 12.

1 Relationship between volume and pressure

The volume of a gas is always the volume of its container as the particles spread out fully. Its pressure is the result of the particles colliding with the walls of the container, and will increase when the frequency or energy of these collisions increases.

If the temperature is held constant, it is found that increasing the pressure on a fixed mass of gas decreases its volume. In other words, the pressure of a gas is inversely proportional to its volume, and the product of pressure and volume is a constant.

TOK

The ideal gas equation can be deduced from a small number of assumptions of ideal behaviour. What is the role of reason, perception, intuition, and imagination in the development of scientific models?

The so-called mentos-soda fountain reaction. Dissolved carbon dioxide in the soda changes quickly into gas in the presence of the candy mint mentos. The sudden pressure change causes the soda to be ejected in a fountain of foam.

Different countries continue to use a variety of units for pressure, including millimetres of mercury (mm Hg), torr, pounds per square inch (psi), and atmosphere (atm).

In all the examples here, use of the term 'pressure' assumes 'absolute pressure', which is referenced against a perfect vacuum. This is in contrast to the gauge pressure, which is measured relative to atmospheric pressure.

$P \propto \frac{1}{V}$

PV = a constant

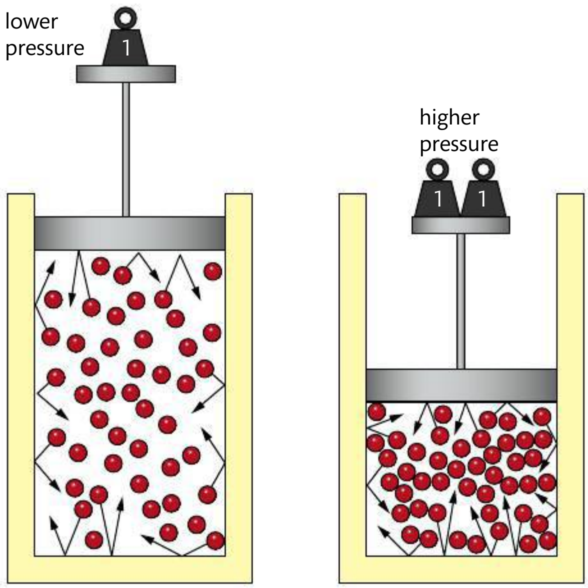


Figure 1.14 As the pressure on a gas is increased, its volume decreases proportionately.

$$P \propto \frac{1}{V}$$

$$PV = \text{a constant}$$

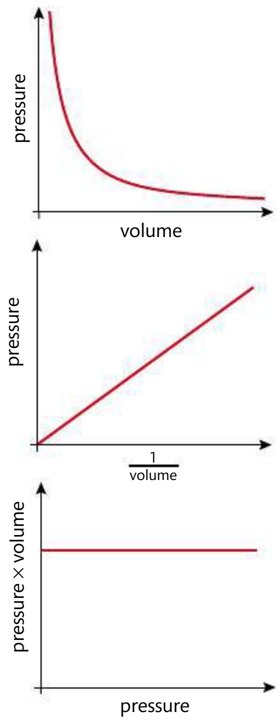


Figure 1.15 Boyle's law. Gas pressure is inversely proportional to its volume.

This relationship is often known as **Boyle's law**, as it was first established by Robert Boyle in 1662. Application of this relationship is found in the compression of gases under pressure, often useful in transport and storage.

2 Relationship between volume and temperature

An increase in temperature represents an increase in the average kinetic energy of the particles. If the pressure is held constant, it is found that increasing the temperature of a fixed mass of gas increases its volume. In other words, the volume of a gas is directly proportional to its absolute temperature, and volume divided by absolute temperature is a constant.

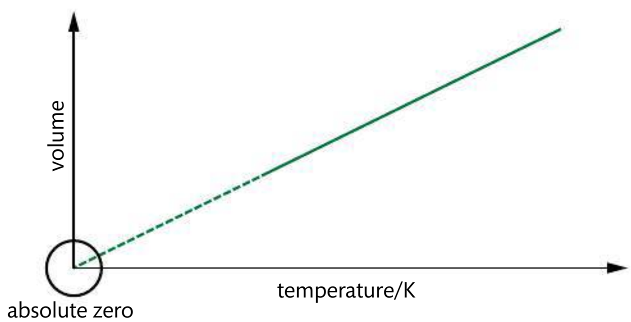


Figure 1.16 Charles' law. Gas volume is proportional to the absolute temperature. Note the dotted line represents an extrapolation, as data at temperatures down to 0 K are not obtainable.

$$V \propto T$$

$$\frac{V}{T} = \text{a constant}$$

$V \propto T$

$\frac{V}{T}$ = a constant

TOK

The SI base unit of temperature is the Kelvin, which has a natural basis for its definition of zero. By contrast, zero in the imperial scale of degree Celsius is arbitrarily defined. What are the implications of using a scale based on natural or arbitrary values?

This relationship is often known as **Charles' law**, as it was first established by Frenchman Jacques Charles in the late 18th century. You can demonstrate this relationship by immersing dented table tennis balls in warm water. As the air inside the ball equilibrates to the temperature of the water, it expands, pushing the dents out on the surface.

Balloons dipped in a mixture of dry ice (solid CO_2) and water rapidly deflate as the air inside contracts at low temperature.

3 Relationship between pressure and temperature

An increase in temperature increases the average kinetic energy of the particles. The particles move faster and collide with the walls of the container with more energy and more frequency, raising the pressure. If the volume is held constant, it is found that increasing the temperature of a fixed mass of gas proportionately raises its pressure. In other words, the pressure of a gas is directly proportional to the absolute temperature, and pressure divided by temperature is a constant.

$$P \propto T$$

$$\frac{P}{T} = \text{a constant}$$

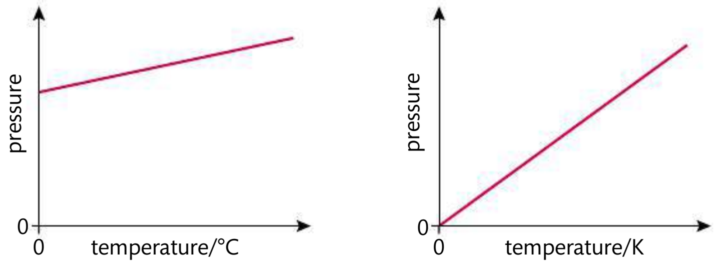


Figure 1.17 Gas pressure is proportional to the absolute temperature.

Always be certain to use temperature in Kelvin only when applying these relationships.

The label on the aerosol can warns of the dangers of exposing the pressurized contents to high temperature.

$$P \propto T$$

$$\frac{P}{T} = \text{a constant}$$

NATURE OF SCIENCE

Scientists work to contribute to a common body of knowledge. Results are shared through publication, considered, and used as the basis for further studies by other scientists. Although many scientific discoveries, such as the gas laws, are named after the key scientist involved, in reality the theories and laws of science stand apart from the individual discoverers. Without Charles and Boyle, the relationships between pressure, volume, and temperature of a gas would still exist. This is very different from the arts. Without Shakespeare, there would have been no *Hamlet*, without Picasso, no *Guernica*.

Pressurized cans, such as soda or beer, often carry a warning to be stored in a cool place. This is because at higher temperatures the pressure inside the can at fixed volume is able to rise to the point of causing the can to explode.

These three gas laws applied to a fixed mass of gas can be summarized as follows:

$$P \propto \frac{1}{V} \textbf{ at constant temperature}$$

$$V \propto T \textbf{ at constant pressure}$$

$$P \propto T \textbf{ at constant volume}$$

These can be combined to give one equation for a fixed mass of gas:

$$\frac{PV}{T} = \textbf{a constant} \text{ or}$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$ (where 1 and 2 refer to initial and final conditions respectively)

Application of this enables gas volume, pressure, and temperature to be calculated as conditions change.

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Worked example

What happens to the volume of a fixed mass of gas when its pressure and its absolute temperature are both doubled?

Solution

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$P_2 = 2 \times P_1$ and $T_2 = 2 \times T_1$, so these can be substituted into the equation:

$$\frac{P_1V_1}{T_1} = \frac{2P_1V_2}{2T_1}$$

We can cancel P_1 and T_1 from both sides and 2s on the right side, leaving

$$V_1 = V_2$$

The volume does not change.

Note that temperature must be in Kelvin in these calculations. However, the units of volume and pressure are not specified, so long as they are consistent on both sides of the equation.

Worked example

The molar volume of a gas at STP is 22.7 $dm^3\ mol^{-1}$. Calculate the molar volume at 25 °C at the same pressure.

Solution

As the pressure is not changing, we do not need to insert P_1 and P_2 into the combined gas equation. Temperature must be converted from °C to K.

$$T_1 = 273\text{ K}, T_2 = 25 + 273 = 298\text{ K}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{22.7\text{ dm}^3}{273\text{ K}} = \frac{V_2}{298\text{ K}}$$

$$V_2 = \frac{298 \times 22.7\text{ dm}^3}{273} = 24.8\text{ dm}^3$$

The ideal gas equation is derived from the combined gas equation and Avogadro's law

The combined gas equation tells us that

$$\frac{PV}{T} = \text{a constant}$$

The value of the constant is directly proportional to the fixed mass of gas, or the number of moles, n.

So $\frac{PV}{T} \propto n$

This can be made into an equation by introducing a constant, $\boldsymbol{R}$, known as the **universal gas constant**.

$\therefore \frac{PV}{T} = nR$, which is usually written as $\boldsymbol{PV = nRT}$

This equation is known as the **ideal gas equation**, and is given in Section 1 of the IB data booklet. The value of R can be calculated by substituting known values into the equation, such as those for the molar volume of a gas at STP. In this case:

$$P = 10^5 \text{ Pa (N m}^{-2}\text{)},\ V = 2.27 \times 10^{-2} \text{ m}^3,\ T = 273 \text{ K},\ n = 1$$

$$\therefore 10^5 \text{ N m}^{-2} \times 2.27 \times 10^{-2} \text{ m}^3 = 1 \text{ mol} \times R \times 273 \text{ K}$$

$$R = 8.31 \text{ N m K}^{-1} \text{ mol}^{-1} \text{ or } 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

This value for R, the gas constant, is given in Section 2 in the IB data booklet and should be used for all calculations involving the ideal gas equation.

Use of the ideal gas equation enables us to calculate how systems respond to changes in pressure, volume, and temperature, and to calculate molar mass. Gas density can also be derived by applying the relationship density = $\frac{\text{mass}}{\text{volume}}$.

These calculations usually involve simply substituting values into the equation, but the use of units needs special attention here. The guidelines below, based on the use of SI units only, should help you avoid some of the common mistakes that arise.

- Pressure, P: must be in Pa (N m^{-2}); if kPa are given, multiply by 10^3.
- Volume, V: must be in m^3; if dm^3 are given, divide by 10^3, if cm^3 are given divide by 10^6.
- Number of moles, n: this is often derived by application of $n = \frac{m}{M}$.
- Temperature, T: must be in Kelvin; if °C is given, add 273.15.

The ideal gas equation is *PV = nRT*

Many sources give data for ideal gas law questions in non-SI units such as atmosphere, which require a mathematical conversion. You will not be expected to be familiar with these conversions in IB examinations.

Worked example

A helium party balloon has a volume of 18.0 dm^3. At 25 °C the internal pressure is 108 kPa. Calculate the mass of helium in the balloon.

Solution

First ensure all data are in SI units:

$$P = 108 \text{ kPa} = 108 \times 10^3 \text{ Pa}$$

$$V = 18.0 \text{ dm}^3 = 18.0 \times 10^{-3} \text{ m}^3$$

$$T = 25 \text{ °C} = 298 \text{ K}$$

As this example shows, if pressure is given in kPa *and* volume is given in dm^3, the same answer is obtained if these values are used directly in the ideal gas equation in place of Pa and m^3.

$$PV = nRT$$

$$108 \times 10^3\ \text{Pa} \times 18.0 \times 10^{-3}\ \text{m}^3 = n \times 8.31\ \text{J K}^{-1}\ \text{mol}^{-1} \times 298\ \text{K}$$

$$\therefore n(\text{He}) = 0.785\ \text{mol}$$

$$\therefore \text{mass (He)} = n\,M = 0.785\ \text{mol} \times 4.00\ \text{g mol}^{-1} = 3.14\ \text{g}$$

Worked example

A sample of gas has a volume of 445 cm^3 and a mass of 1.500 g at a pressure of 95 kPa and a temperature of 28 °C. Calculate its molar mass.

Solution

Substitute $n = \frac{m}{M}$ into the ideal gas equation, and rearrange to solve for M.

$$M = \frac{mRT}{PV}$$

Ensure all data are in SI units:

$$P = 95\ \text{kPa} = 95 \times 10^3\ \text{Pa}$$

$$V = 445\ \text{cm}^3 = 445 \times 10^{-6}\ \text{m}^3$$

$$T = 28\ °\text{C} = 301\ \text{K}$$

$$\therefore M = \frac{1.500\ \text{g} \times 8.31\ \text{J K}^{-1}\ \text{mol}^{-1} \times 301\ \text{K}}{95 \times 10^3\ \text{Pa} \times 445 \times 10^{-6}\ \text{m}^3} = 88.8\ \text{g mol}^{-1}$$

Worked example

A gas has a density of 1.65 g dm^{-3} at 27 °C and 92.0 kPa. Determine its molar mass.

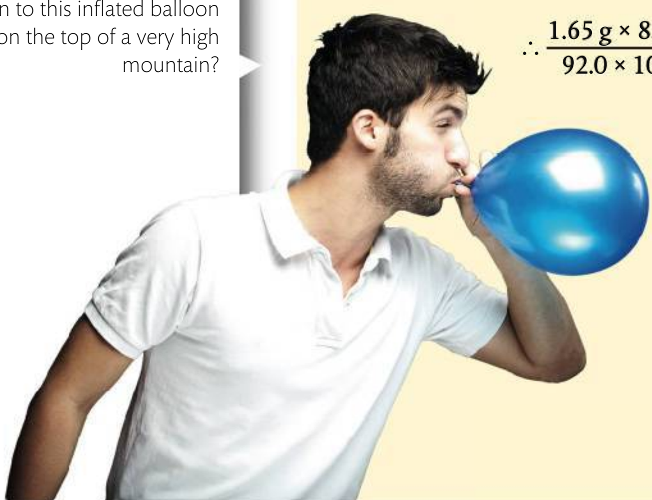
Solution

$$\text{density data} \Rightarrow 1.65\ \text{g occupies}\ 1.00\ \text{dm}^3$$

As in the example above, substitute data with correct units into $M = \frac{mRT}{PV}$

$$\therefore \frac{1.65\ \text{g} \times 8.31\ \text{J K}^{-1}\ \text{mol}^{-1} \times 300\ \text{K}}{92.0 \times 10^3\ \text{Pa} \times 1.00 \times 10^{-3}\ \text{m}^3} = 44.7\ \text{g mol}^{-1}$$

Blowing up a balloon increases its volume as the number of particles increases. What do you think would happen to this inflated balloon on the top of a very high mountain?



Experiment to calculate the molar mass of carbon dioxide by application of the ideal gas equation

Full details with a worksheet are available online.

A known mass of $CuCO_3(s)$ is heated and the gas evolved collected by displacement of water. The volume of the gas, the room temperature, and pressure are recorded.

$$CuCO_3(s) \rightarrow CuO(s) + CO_2(g)$$

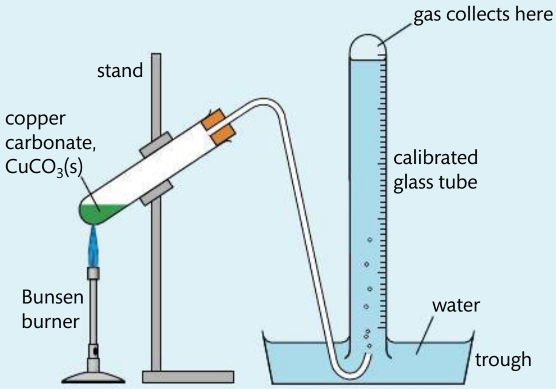


Figure 1.18 Experiment to calculate the molar mass of carbon dioxide.

Sample results

	Trial 1	Observations
mass of boiling tube + $CuCO_3$ before heating / g ± 0.001	33.910	$CuCO_3$ is a green powder
mass of boiling tube + $CuCO_3$ after heating / g ± 0.001	33.822	the contents of the tube are black after heating
mass change (mass CO_2) / g ± 0.002	0.088	
volume of gas collected / cm^3 ± 0.1	38.1	the gas collected is colourless
temperature / K ± 0.1	293.0	
pressure / Pa ± 0.1	101.3 kPa	

Processed data

$$M(CO_2) = \frac{mRT}{PV} = \frac{0.088\ g \times 8.31\ J\ mol^{-1}\ K^{-1} \times 293.0\ K}{101.3\ kPa \times 0.0381\ dm^3}$$

experimental value $M(CO_2) = 55.5\ g\ mol^{-1}$

theoretical value $M(CO_2) = 44.01\ g\ mol^{-1}$

$\therefore$ % error = 26.1%

The percentage error can be analysed in consideration of systematic errors such as:

- gas collected may not be pure CO_2;
- CO_2 may be soluble in water;
- air in the tube is collected with the gas;
- gas collected has not equilibrated to room temperature.

Modifications to the experimental design should suggest ways to reduce the impact of these errors. Note that repeat trials and error propagation are not shown here.

NATURE OF SCIENCE

Scientists often use models, which may be simple or complex, to explain concepts that are not observable. By their nature, models are thinking tools and so they all have limitations. Models are often tested against experimental results, and should be able to explain phenomena that are seemingly different from the ones used to develop the model. Sometimes, when a model does not seem consistent with data or observations, it may need to be modified or replaced. At other times it can be retained, with limitations to its usefulness agreed.

The ideal gas model has all of these characteristics. While it provides a useful conceptual image of gas behaviour, and is consistent with some testable predictions, there are limitations to this, as we will see below. These do not mean, however, that the model should be discarded – rather they alert us to the fact that the model is in some ways an over-simplification, and cannot apply accurately under all conditions.

Exercises

51 A 2.50 dm^3 container of helium at a pressure of 85 kPa was heated from 25 °C to 75 °C. The volume of the container expanded to 2.75 dm^3. What was the final pressure of the helium?

52 After a sample of nitrogen with a volume of 675 cm^3 and a pressure of 1.00×10^5 Pa was compressed to a volume of 350 cm^3 and a pressure of 2.00×10^5 Pa, its temperature was 27.0 °C. Determine its initial temperature.

53 The absolute temperature of 4.0 dm^3 of hydrogen gas is increased by a factor of three and the pressure is increased by a factor of four. Deduce the final volume of the gas.

54 To find the volume of a flask, it was first evacuated so that it contained no gas at all. When 4.40 g of carbon dioxide was introduced, it exerted a pressure of 90 kPa at 27 °C. Determine the volume of the flask.

55 An unknown noble gas has a density of 5.84 g dm^{-3} at STP. Calculate its molar mass, and so identify the gas.

56 A 12.1 mg sample of a gas has a volume of 255 cm^3 at a temperature of 25.0 °C and a pressure of 1300 Pa. Determine its molar mass.

57 Which has the greater density at STP, hydrogen or helium?

58 Calculate the volume of oxygen at STP required for the complete combustion of 125 g of octane, C_8H_{18}, to form carbon dioxide and water.

59 A sample of an unknown gas with a mass of 3.620 g occupied a volume of 1120 cm^3 at a pressure of 99 kPa and a temperature of 25.0 °C. The sample contained 2.172 g O and 1.448 g S. Determine the empirical and molecular fomula of the gas.

60 A road cyclist pumps his tyres up very hard before a trip over a high mountain pass at high altitude. Near the summit one of his tyres explodes. Suggest why this may have occurred.

Real gases show deviation from ideal behaviour

An ideal gas is defined as one that obeys the ideal gas law $PV = nRT$ under all conditions. This means that for one mole of gas, the relationship PV/RT should be equal to 1. So a graph of PV/RT against P for one mole of an ideal gas is a horizontal line of intercept 1 (Figure 1.19).

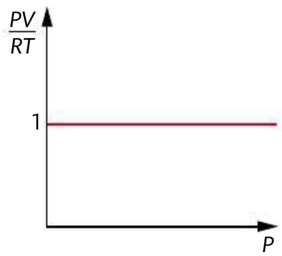


Figure 1.19 For one mole of an ideal gas, the relationship *PV/RT* is a constant at all pressures.

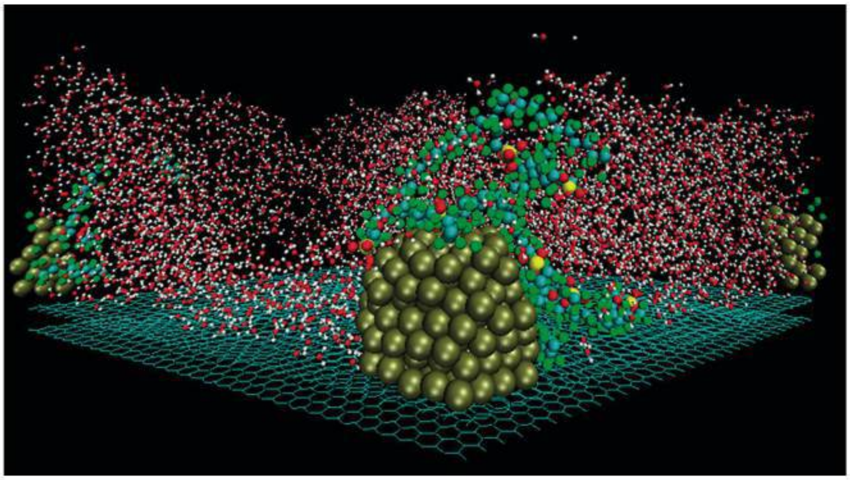
But, as we noted earlier, there is no such thing as an ideal gas. All gases, known as **real gases**, deviate to some extent from ideal behaviour. So, for real gases the value of PV/RT for one mole will vary. An example of the extent of this variation from 1 at different conditions is shown in Figure 1.20.

06 Chemical kinetics

Copper is commonly used in roofing due to its durability; however, over time it reacts with oxygen and turns green. Scientists have found ways to both accelerate and slow down this reaction in order to get the desired appearance of the metal.

Essential ideas

6.1	The greater the probability that molecules will collide with sufficient energy and proper orientation, the higher the rate of reaction.
16.1	Rate expressions can only be determined empirically and these limit possible reaction mechanisms. In particular cases, such as a linear chain of elementary reactions, no equilibria, and only one significant activation barrier, the rate equation is equivalent to the slowest step of the reaction.
16.2	The activation energy of a reaction can be determined from the effect of temperature on reaction rate.



Computer simulation of fuel cell nanocatalysts. Platinum nanoparticles, shown in yellow, on a carbon substrate, shown as the green grid, are being investigated for their catalytic activity in fuel cell reactions. The red and white molecules are water, and the coloured chains are fragments of a polymer used in the proton exchange membranes. Research and development in fuel cell technology rely on an understanding of the factors that control reaction rates, including the action of catalysts.

The word **kinetics**, derived from the Greek word *kinesis*, refers to movement. The word 'cinema' (kine-ma) has the same origin and is used to describe 'the movies'. Movement in chemistry refers to the progress of a reaction, so kinetics is the study of how fast a reaction goes.

Imagine you are cooking in the kitchen. As you drop an egg into hot fat in the pan it immediately changes to a white solid; meanwhile a container of milk that was left out of the refrigerator is slowly turning sour. We observe a similar wide variation in the rate of reactions that we study in the laboratory, and these data can be very useful. Kinetic studies are of prime importance in industry because they give information on how quickly products form and on the conditions that give the most efficient and economic yield. They can also be useful in situations where we want to slow reactions down – for example, those that cause the destruction of ozone in the stratosphere, or the reactions where pollutants in the air combine to produce smog. At other times, it is important to know for how long a certain reaction will continue – for example, the radioactive effect from radioactive waste. Knowledge of reaction kinetics also gives insights into how reactions happen at the molecular level by suggesting a sequence of bond breaking and bond making, known as the **reaction mechanism**.

This chapter begins with a study of reaction rates and a consideration of how these are measured in different cases. Through the postulates of the **collision theory**, we will come to understand why different factors affect the rate of reactions, and then go on to develop mathematical equations to explore these relationships.

6.1 Collision theory and rates of reaction

Understandings:

- Species react as a result of collisions of sufficient energy and proper orientation.
- The rate of reaction is expressed as the change in concentration of a particular reactant/product per unit time.
- Concentration changes in a reaction can be followed indirectly by monitoring changes in mass, volume, and colour.
- Activation energy (E_a) is the minimum energy that colliding molecules need in order to have successful collisions leading to a reaction.
- By decreasing E_a, a catalyst increases the rate of a chemical reaction, without itself being permanently chemically changed.

Applications and skills:

- Description of the kinetic theory in terms of the movement of particles whose average kinetic energy is proportional to temperature in Kelvin.
- Analysis of graphical and numerical data from rate experiments.

Guidance

- *Calculation of reaction rates from tangents of graphs of concentration, volume, or mass vs time should be covered.*
- *Students should be familiar with the interpretation of graphs of changes in concentration, volume, or mass against time.*

- Explanation of the effects of temperature, pressure/concentration, and particle size on rate of reaction.
- Construction of Maxwell–Boltzmann energy distribution curves to account for the probability of successful collisions and factors affecting these, including the effect of a catalyst.
- Investigation of rates of reaction experimentally and evaluation of the results.
- Sketching and explanation of energy profiles with and without catalysts.

Rate of reaction is defined as the rate of change in concentration

When we are interested in how quickly something happens, the factor that we usually measure is **time**. For example, in a sports race the competitors are judged by the time it takes them to reach the finishing line. However, if we want to compare their performance in different races over different distances, we would need to express this as a **rate** – in other words how they performed *per unit time*.

Note that because time and rate are reciprocal values, as one increases, the other decreases. So in the example of a race, the racer with the *shortest time* wins the race, because they had the *highest rate*.

Rate takes the reciprocal value of time, so is expressed *per unit time* or in SI units *per second* (symbol = s^{-1}).

$$\text{rate} = \frac{1}{\text{time}} = \frac{1}{\text{s}} = \text{s}^{-1}$$

In the study of chemical reactions we use the concept of **rate of reaction** to describe how quickly a reaction happens. As the reaction proceeds, reactants are converted into products, and so the concentration of reactants decreases as the concentration of products increases. The graphs in Figures 6.1 and 6.2 show sketches of typical data from reactions.

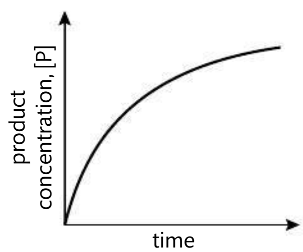


Figure 6.1 Concentration of product against time.

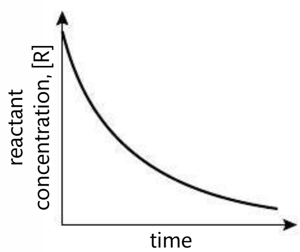


Figure 6.2 Concentration of reactant against time.

The rate of a reaction depends on how quickly the concentration of either reactant or product changes with respect to time. It can be defined as follows:

$$\text{rate of reaction} = \frac{\text{increase in product conc.}}{\text{time taken}} \text{ or } \frac{\text{decrease in reactant conc.}}{\text{time taken}}$$

Using Δ to represent 'change in', [R] for concentration of reactant, and [P] for concentration of product, we can express this as

$$\text{rate of reaction} = \frac{\Delta[P]}{\Delta t} \text{ or } -\frac{\Delta[R]}{\Delta t}$$

The negative sign in the reactant expression indicates that reaction concentration is decreasing, but, by convention, rate is expressed as a positive value.

As rate = change in concentration per time, its units are **mol dm^{-3} s^{-1}**.

Figure 6.3 represents graphs of two different reactions showing the change in concentration of reactant against time. We can see that the concentration of reactants is decreasing more quickly in reaction A than in reaction B – the curve is steeper. The steepness, or gradient, of the curve is a measure of the change in concentration per unit time, in other words, the rate of the reaction.

Because the graphs are curves and not straight lines, the gradient is not constant and so can only be given for a particular value of time. We can measure this by drawing a tangent to the curve at the specified time point and measuring its gradient. This is shown in Figure 6.4 for the time point 120 seconds. Note that the gradient of this graph is negative as the reactant concentration is decreasing, but rate is expressed as a positive value.

The rate of a chemical reaction is the increase in concentration of products or the decrease in concentration of reactants per unit time.

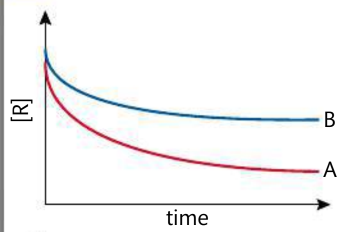


Figure 6.3 Reactant concentration against time for two different reactions A and B.

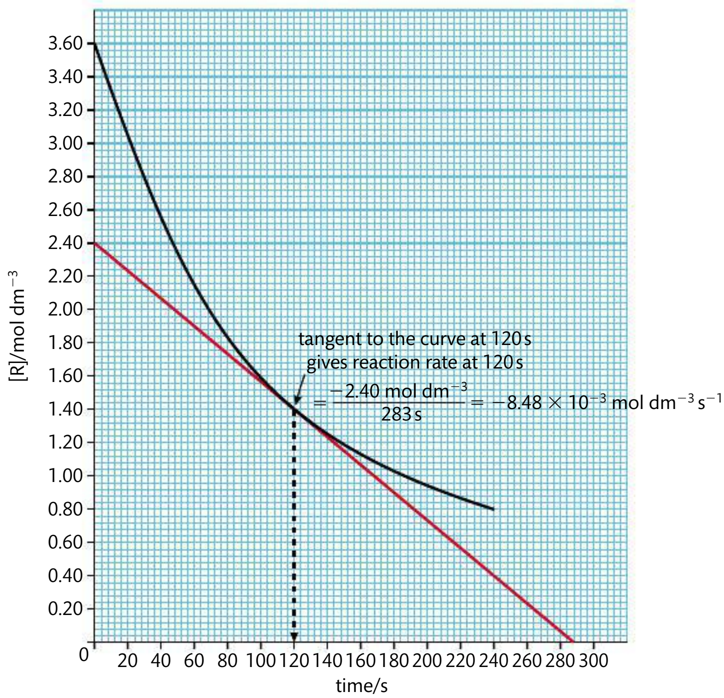


Figure 6.4 Measuring the gradient of the tangent to the curve at time t = 120 seconds. The measured rate of the reaction is 8.48×10^{-3} mol dm^{-3} s^{-1} at this time.

Note the difference between the two instructions 'draw a graph' and 'sketch a graph'. *Drawing* is based on actual data so scales must be chosen appropriately and data points and units clearly marked; *sketching* is a way of showing a trend in the variables, without reference to specific data. Note that in both cases the axes must be clearly labelled with the dependent variable on the *y*-axis and the independent variable on the *x*-axis.

The shape of the curve indicates that the rate of the reaction is not constant during the reaction, but is greatest at the start and slows down as the reaction proceeds. This is illustrated in Figure 6.5, which shows the calculation of reaction rate at two different times during a reaction.

- $t = 0$ s rate = 2.9×10^{-2} mol dm^{-3} s^{-1}
- $t = 90$ s rate = 7.2×10^{-3} mol dm^{-3} s^{-1}

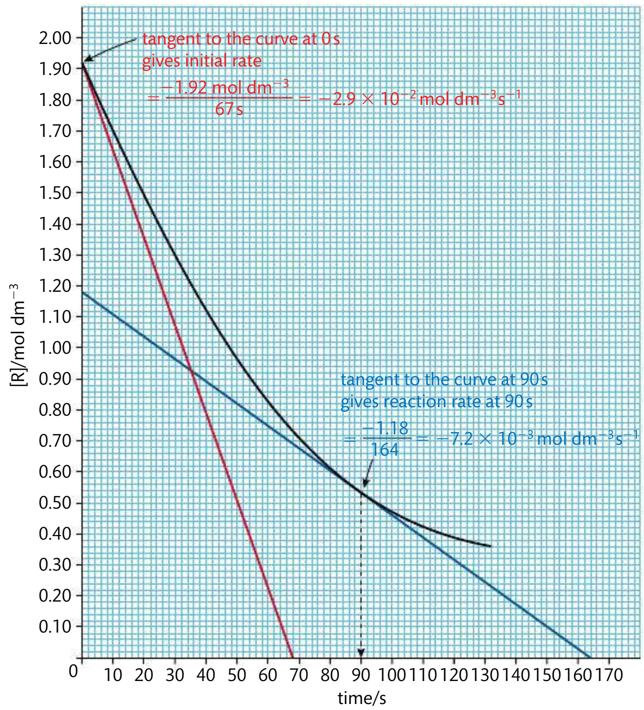


Figure 6.5 Measuring the gradient of the tangent to the curve at two different times during a reaction. The rate decreases as the reaction proceeds.

You will get a more accurate value for the slope of the tangent if you draw it as long as you reasonably can so that the 'y' value (concentration) and the 'x' value (time) are as large as possible.

The rate of a reaction is not constant during a reaction, but is greatest at the start and decreases as the reaction proceeds.

The rate is greatest at the start because this is when the reactant concentration is highest. The effect of concentration on reaction rate is discussed later in this chapter. Because of this variation in rate as a reaction proceeds, in order to compare the rates of reactions under different conditions it is common to compare the **initial rate** of each reaction by taking the tangent to the curve at $t = 0$. As we will see later in this chapter, initial rates data are very useful in analysing the effect of concentration on rate.

Measuring rates of reaction uses different techniques depending on the reaction

Choosing whether to measure the change in concentration of reactants or products, and the technique with which to measure that change really depends on what is the most convenient for a particular reaction. This is different for different reactions. In most cases the concentration is not measured directly, but is measured by means of a signal which is related to the changing concentration. If, for example, a reaction produces a coloured precipitate as product, the change in colour could be measured; if a reaction gives off a gas, then the change in volume or change in mass could be measured. The raw data collected using these 'signals' will be in a variety of units, rather than as concentrations measured in mol dm^{-3} directly. This is generally not a problem, as it still enables us to determine the rate of the reaction. Many of the experiments used here can be carried out using data-logging devices. Some examples of common techniques in relation to specific reactions are discussed below.

TOK

Note that the balanced equation of a reaction gives us no information about its rate. We can obtain this information only from experimental (empirical) data. Is there a fundamental difference between knowledge claims based on conclusions from theoretical data and those based on experimental data?

1 Change in volume of gas produced

This is a convenient method if one of the products is a gas. Collecting the gas and measuring the change in volume at regular time intervals enables a graph to be plotted of volume against time. A **gas syringe** can be used for this purpose. It consists of a ground glass barrel and plunger, which moves outwards as the gas collects and is calibrated to record the volume directly. Alternatively, the gas can be collected by displacement of water from an inverted burette. Note though that the displacement method can only be used if the gas collected has low solubility in water. Data loggers are also available for continuous monitoring of volume change against time. The rate of reaction of a metal with dilute acid to release hydrogen gas can be followed in this way.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

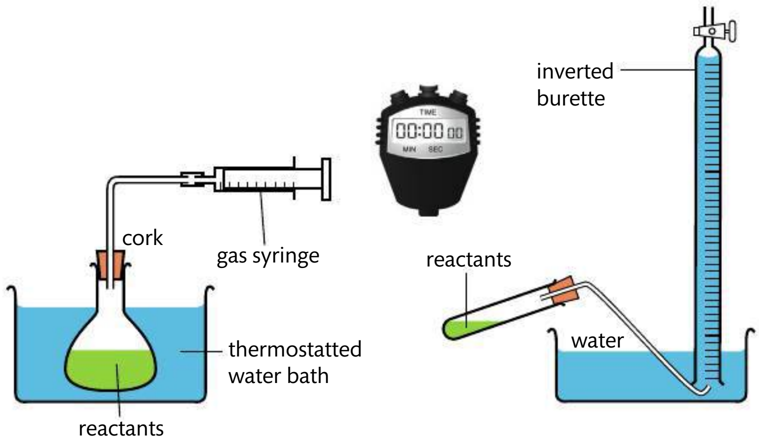
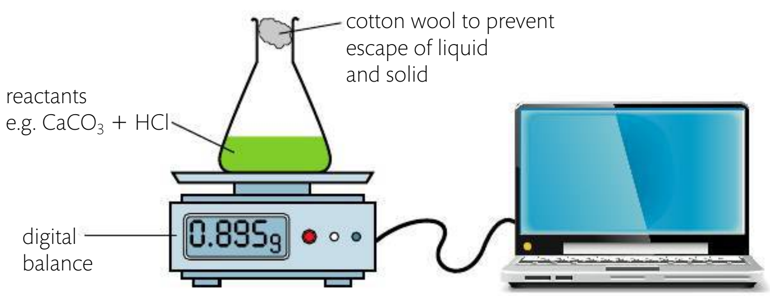


Figure 6.6 Experiments to measure rate of reaction by following change in volume against time.

2 Change in mass

Many reactions involve a change in mass, and it may be convenient to measure this directly. If the reaction is giving off a gas, the corresponding decrease in mass can be measured by standing the reaction mixture directly on a balance. This method is unlikely to work well where the evolved gas is hydrogen, as it is too light to give significant changes in mass. The method allows for continuous readings, so a graph can be plotted directly of mass against time. For example, the release of carbon dioxide from the reaction between a carbonate and dilute acid can be followed in this way.

$$CaCO_3(s) + 2HCl\ (aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$



CHALLENGE YOURSELF

1 Most gases are less soluble in warm water than in cold water, so using warm water may help to reduce a source of error when collecting a gas by displacement of water. But what new error might this create?

When you take a glass of cold water from the refrigerator and leave it by your bed overnight, you may notice bubbles of gas have formed by morning. This is because the gas (mostly dissolved oxygen) has become less soluble as the temperature of the water has increased.

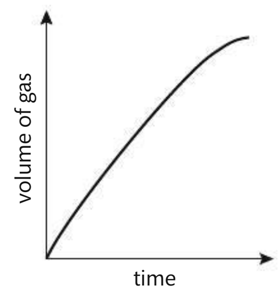


Figure 6.7 Volume of gas against time.

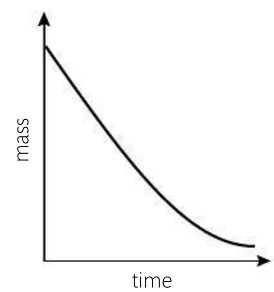


Figure 6.9 Mass against time.

Figure 6.8 Experiment to measure rate of reaction by following change in mass against time.

3 Change in transmission of light: colorimetry/spectrophotometry

This technique can be used if one of the reactants or products is coloured and so gives characteristic absorption in the visible region (wavelengths about 320–800 nm). Sometimes an indicator can be added to generate a coloured compound that can then be followed in the reaction. A colorimeter or spectrophotometer works by passing light of a selected wavelength through the solution being studied and measures the intensity of the light transmitted by the reaction components. As the concentration of the coloured compound increases, it absorbs proportionally more light, so less is transmitted. A photocell generates an electric current according to the amount of light transmitted and this is recorded on a meter or connected to a computer.

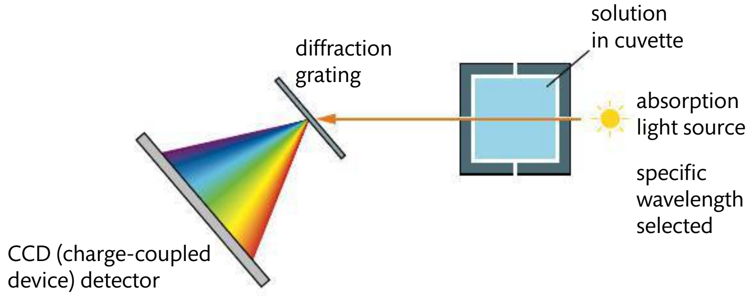


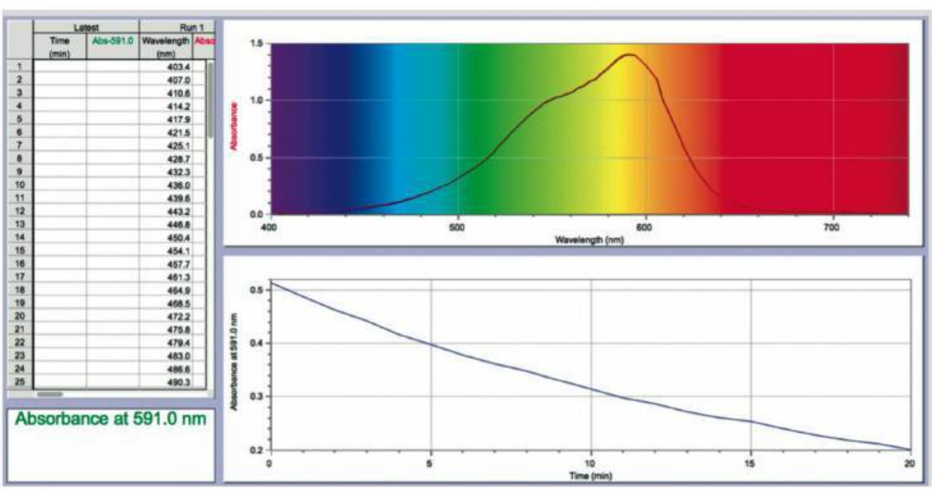
Figure 6.10 A spectrophotometer measures absorbance, which directly relates to concentration. Light of a specific wavelength is selected from an internal source and passes through the solution placed in the cuvette. The transmitted light passes through a diffraction grating, then the diffracted light is collected and changed into a digital signal by the CCD detector.

The method allows for continuous readings to be made, so a graph of absorbance against time can be plotted directly. It is possible to convert the absorbance values into concentrations using a standard curve based on readings of known concentration. Often, however, it is sufficient to record absorbance itself (or transmittance, which is inversely proportional) as a function against time.

For example, the reaction between the dye crystal violet and sodium hydroxide solution can be written in an abbreviated form as follows:

$$CV^{+} + OH^{-} \rightarrow CVOH$$

The initial crystal violet solution (CV^{+}) is coloured but this slowly changes to a colourless solution as the product (CVOH) is formed. So we can determine the rate of product formation by measuring the *decrease in absorbance* that occurs as the coloured reactant is depleted. The screen shot shows data from this reaction, with absorbance at 591 nm measured against time.



Results of an experiment to determine the rate of reaction by following change in absorbance against time using a digital spectrophotometer.

An example of the use of absorbance data to follow concentration changes is given in section 16.1 of this chapter. In Chapter 13, Biochemistry, we discuss the use of a calibration curve to determine concentration from absorbance data using the Beer–Lambert law.

4 Change in concentration measured using titration

In some reactions it may be possible to measure the concentration of one of the reactants or products by titrating it against a known 'standard' (see Chapter 1). However, because this technique involves chemically changing the reaction mixture, it cannot be done continuously as the reaction proceeds. Instead, samples must be withdrawn from the reaction mixture at regular time intervals and then analysed by titration. A problem here is that the process of titration takes time, during which the reaction mixture in the sample will continue to react. To overcome this, a technique known as **quenching** can be used, where a substance is introduced which effectively stops the reaction in the sample at the moment it is withdrawn. It is rather like obtaining a 'freeze frame' shot of the reaction at a particular interval of time. In order to see how concentration changes as the reaction proceeds, it is necessary to repeat this process at several intervals of time.

For example, the reaction between H_2O_2 and acidified KI yields I_2, which can be titrated against sodium thiosulfate, $Na_2S_2O_3$ to determine its concentration. Sodium carbonate, Na_2CO_3, is used to quench the reaction by neutralising the added acid.

$$H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \rightarrow I_2(aq) + 2H_2O(l)$$

In all these reactions the goal is to measure change in concentration against time. This allows a comparison of reaction rates under different conditions. But because the rate is dependent on temperature, it is essential to control the temperature throughout these experiments. It is sometimes suitable to carry out the reactions in a thermostatically controlled water bath.

5 Change in concentration measured using conductivity

The total electrical conductivity of a solution depends on the total concentration of its ions and on their charges. If this changes when reactants are converted to products, it can provide a convenient method to follow the progress of the reaction. Conductivity can be measured directly using a conductivity meter which involves immersing inert electrodes in the solution. As with colorimetry, the apparatus can be calibrated using solutions of known concentration so that readings can be converted into the concentrations of the ions present.

For example, in the reaction:

$$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$$

the sharp decrease in the concentration of ions (12 on the reactants side and 0 on the products side) will give a corresponding decrease in the electrical conductivity of the solution as the reaction proceeds.

6 Non-continuous methods of detecting change during a reaction: 'clock reactions'

Sometimes it is difficult to record the continuous change in the rate of a reaction. In these cases, it may be more convenient to measure the time it takes for a reaction to reach a certain chosen fixed point – that is, something observable which can be used as an arbitrary 'end point' by which to stop the clock. The time taken to reach this point for the same reaction under different conditions can then be compared and used as a means of judging the different rates of the reaction. Note though the limitation of this method is that the data obtained give only an average rate over the time interval.

Experiments can be done to investigate the rate of reaction with changing reactant concentration, termperature, particle size, and catalyst. Full details of how to carry out an experiment on the effect of concentration, with a worksheet are available online.

Recording the increase in opaqueness during a reaction. Sodium thiosulfate (left) is a clear liquid, which reacts with hydrochloric acid (upper centre) to form an opaque solution (right). A cross (left) drawn onto paper is placed under the reaction beaker to allow the end point to be confirmed. The experiment is timed from when the reactants are mixed until the cross disappears from sight.

For example, the following can be measured:

- the time taken for a certain size piece of magnesium ribbon to react completely with dilute acid, until it is no longer visible

$$Mg(s) + 2HCl\,(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

- the time taken for a solution of sodium thiosulfate with dilute acid to become opaque by the precipitation of sulfur, so that a cross viewed through the solution is no longer visible

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + SO_2(aq) + H_2O(l) + S(s)$$

Exercises

1 Consider the following reaction:

$$2MnO_4^-(aq) + 5C_2O_4^{2-}(aq) + 16H^+(aq) \rightarrow 2Mn^{2+}(aq) + 10CO_2(g) + 8H_2O(l)$$

Describe three different ways in which you could measure the rate of this reaction.

2 Which units are used to express the rate of a reaction?

A mol dm^{-3} time **B** mol^{-1} dm^3 time^{-1} **C** mol dm^{-3} time^{-1} **D** mol time^{-1}

3 The reaction between calcium carbonate and hydrochloric acid is carried out in an open flask. Measurements are made to determine the rate of the reaction.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

(a) Suggest three different types of data that could be collected to measure the rate of this reaction.
(b) Explain how you would expect the rate of the reaction to change with time and why.

4 The following data were collected for the reaction

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g) \text{ at } 390\,°C$$

$[H_2O_2]$ / mol dm^{-3}	Time / s	$[H_2O_2]$ / mol dm^{-3}	Time / s
0.200	0	0.070	120
0.153	20	0.063	140
0.124	40	0.058	160
0.104	60	0.053	180
0.090	80	0.049	200
0.079	100		

Draw a graph of concentration against time and determine the reaction rate after 60 s and after 120 s.

TOK

The Kelvin scale of temperature gives a natural measure of the kinetic energy of gas, whereas the Celsius scale is based on the properties of water. Are physical properties such as temperature invented or discovered?

Collision theory

Kinetic energy and temperature

Since the early 18th century, theories have been developed to explain the fact that gases exert a pressure. These theories developed alongside a growing understanding of the atomic and molecular nature of matter, and were extended to include the behaviour of particles in all states of matter. Today they are summarized as the **kinetic-molecular theory of matter** (see Chapter 1).

The essence of kinetic molecular theory is that particles in a substance move randomly as a result of the **kinetic energy** that they possess. Because of the random nature of these movements and collisions, not all particles in a substance at any one time have the same values of kinetic energy, but will have instead a range of values. A convenient way to describe the kinetic energy of a substance is therefore to take the *average* of these values, and this is directly related to its **absolute temperature** – that is, its temperature measured in Kelvin.

Increasing temperature therefore means an increase in the average kinetic energy of the particles of a substance. As we supply a substance with extra energy through heating it, we raise the average kinetic energy of the particles and so also raise its temperature. When we compare the behaviour of the particles in the three states of matter from solid, through liquid, to gas, the differences are a result of this increase in the average kinetic energy of the particles, as described on page 9.

For a given substance this can be summarized as follows:

Solid Liquid Gas

increasing kinetic energy

→

increasing temperature

Temperature in Kelvin (K) is proportional to the average kinetic energy of the particles in a substance.

There is more heat in an iceberg than in a cup of boiling coffee.

As heat is a form of energy, the total heat in the iceberg will be the sum of all the energy of all its particles. The iceberg's *temperature* is of course much lower than that of the cup of coffee, which means its particles have, on average, much lower kinetic energy. But because there are vastly more particles in the iceberg than the cup of coffee, its *total* energy is greater.

The Maxwell–Boltzmann distribution curve

The fact that particles in a gas at a particular temperature show a range in their values of kinetic energy is expressed by the **Maxwell–Boltzmann distribution curve**.

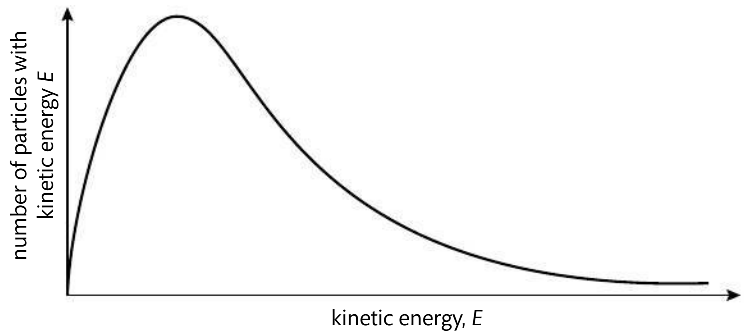


Figure 6.11 The Maxwell–Boltzmann distribution curve.

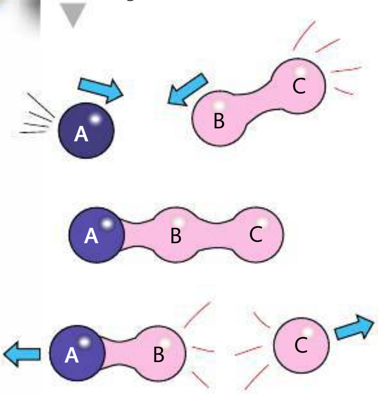
Like other distribution curves, this shows the number having a particular value, in this case of particles with a particular value of kinetic energy (or the probability of that value occurring) plotted against the values for kinetic energy. The area under the curve represents the total number of particles in the sample.

Although their names are linked in the famous energy distribution curve discussed here, James Clerk Maxwell and Ludwig Boltzmann were two people with very different outlooks on life. Maxwell was a Scottish physicist, known for his insatiable curiosity and overflowing humour. His wife worked alongside him in many of his experiments. Boltzmann, an Austrian, was prone to depression and eventually took his own life while on a family holiday, seemingly believing that his work was not valued. Nonetheless as peers during the 19th century, both seeking to explain the observed properties of gases, they refined and developed each other's ideas, culminating in the distribution curve that bears their two names.

How reactions happen

When reactants are placed together, the kinetic energy that their particles possess causes them to collide with each other. The energy of these collisions may result in some bonds between the reactants being broken, and some new bonds forming. As a result, products form and the reaction 'happens'.

Figure 6.12 Particles react by colliding.



It follows that the rate of the reaction will depend on the number of collisions between particles which are 'successful', that is, which lead to the formation of products. But a very important point is that *not all collisions will be successful*. There are two main factors that influence this:

• energy of collision and
• geometry of collision.

i Energy of collision

Activation energy (E_a) is defined as the minimum value of kinetic energy which particles must have before they are able to react.

In order for a collision to lead to reaction, the particles must have a certain minimum value for their kinetic energy, known as the **activation energy**, $\boldsymbol{E_a}$. This energy is necessary for overcoming repulsion between molecules, and often for breaking some bonds in the reactants before they can react. When this energy is supplied, the reactants achieve the **transition state** from which products can form. The activation energy therefore represents an energy barrier for the reaction, and it has a different value in different reactions.

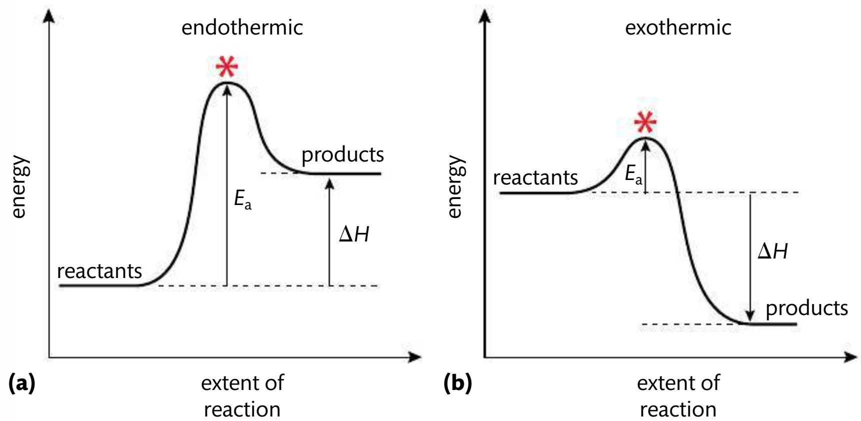


Figure 6.13 Energy path of a reaction with the activation energy for (a) endothermic and (b) exothermic reactions. * represents the transition state.

We can think of the activation energy as a 'threshold value' – a bit like a pass mark in an examination: values greater than this mark achieve a pass, lower values do not achieve a pass. The activation energy threshold similarly determines which particles react and which do not. So only particles that have a kinetic energy value greater than the activation energy will have successful collisions. Note that particles with lower values of kinetic energy may still collide, but these collisions will not be 'successful' in the sense of causing a reaction.

It therefore follows that the rate of the reaction depends on the proportion of particles that have values for kinetic energy greater than the activation energy.

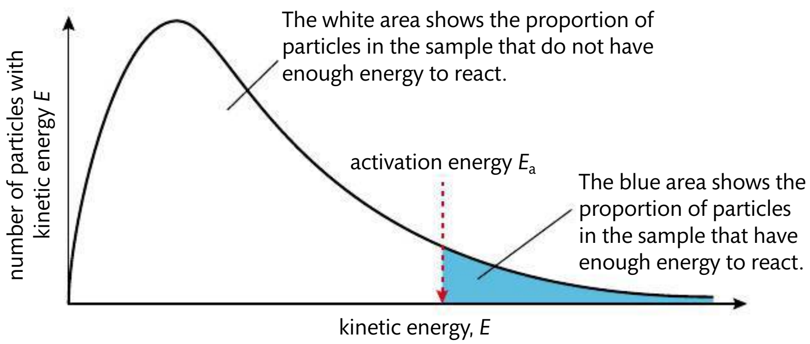


Figure 6.14 Maxwell–Boltzmann distribution showing how the activation energy, E_a, distinguishes between particles which have greater or lesser values of kinetic energy.

The magnitude of the activation energy varies greatly from one reaction to another, and it is an important factor in determining the overall rate of the reaction. In general, reactions with high activation energy proceed more slowly than those with low activation energy. In section 16.2 of this chapter, we will learn how activation energy can be calculated.

ii Geometry of collision

Because collisions between particles are random, they are likely to occur with the particles in many different orientations. In some reactions this can be crucial in determining whether or not the collisions will be successful, and therefore what proportion of collisions will lead to a reaction.

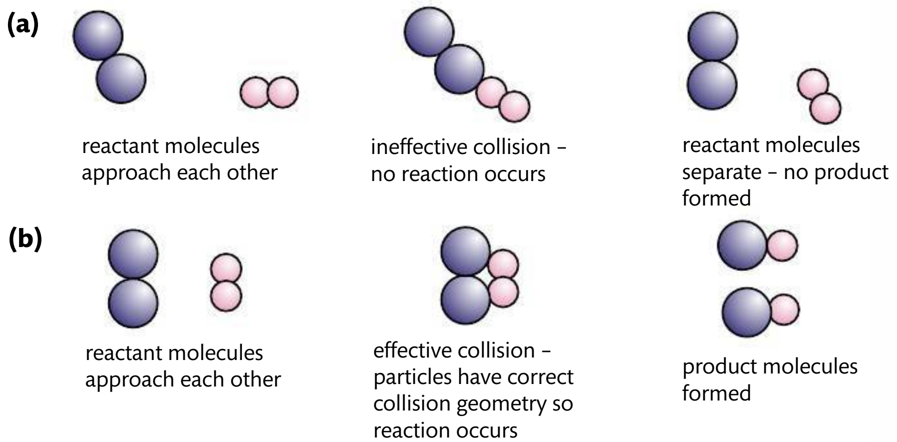


Figure 6.15 The effect of collision geometry. In (a) the particles do not have the correct collision geometry and no reaction occurs. In (b) the particles collide with the correct geometry, enabling products to form.

Summary

The rate of a reaction depends on the frequency of collisions which occur between particles possessing both:

- values of kinetic energy greater than the activation energy and
- appropriate collision geometry.

Understanding this theory will help us to investigate and explain the factors that influence the rate of reaction.

In order to react, particles must collide with kinetic energy greater than the activation energy and have the correct collision geometry.

Exercises

5 Which statement is correct for a collision between reactant particles that leads to reaction?

A Colliding particles must have different energy.
B Colliding particles must have the same energy.
C Colliding particles must have kinetic energy greater than the average kinetic energy.
D Colliding particles must have kinetic energy greater than the activation energy.

6 Which of the following determine whether a reaction will occur?

I the orientation of the molecules
II the energy of the molecules
III the volume of the container

A I and II **B** I and III **C** II only **D** I, II, and III

7 If we compare two reactions, one which requires the simultaneous collision of three molecules and the other which requires a collision between two molecules, which reaction would you expect to be faster? Explain why.

NATURE OF SCIENCE

The collision between a small number of reacting species resulting in chemical change has not been observed directly. Instead, collision theory is based on an application of kinetic molecular theory and theory of how chemical reactions occur through bond breaking and bond making. In science, theories generally accommodate the assumptions and premises of other theories in this way. The fact that collision theory helps to explain the observed effects of factors influencing the rate of reactions adds to its validity. The theory enables chemists to make predictions about the impact of different factors on the rates of specific reactions, which has important applications in many branches of chemistry such as industry, biochemistry, and environmental chemistry.

Be careful not to confuse the question of 'how fast?' a reaction goes with the question of 'how far?' it goes. We are discussing only the first question here. The question of how far a reaction goes, which influences the *yield* of the reaction, will be discussed in Chapter 7, Equilibrium.

Factors affecting rate of reaction

From the collision theory we know that any factor which increases the number of successful collisions will increase the rate of the reaction. We will investigate five such factors here.

1 Temperature

Increasing the temperature causes an increase in the average kinetic energy of the particles. We can see this by comparing Maxwell–Boltzmann distribution curves of the same sample of particles at two different temperatures.

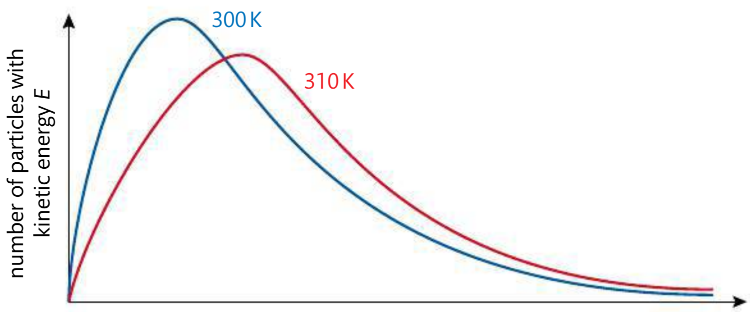


Figure 6.16 Maxwell–Boltzmann distribution curves for a sample of gas at 300 K and 310 K. At the higher temperature the curve has shifted to the right and become broader, showing a larger number of particles with higher values for kinetic energy.

The area under the two curves is equal as this represents the total number of particles in the sample. But at the higher temperature, more of the particles have higher values for kinetic energy and the peak of the curve shifts to the right. In Figure 6.17, we can see how this shift increases the proportion of particles that have values for kinetic energy greater than that of the activation energy.

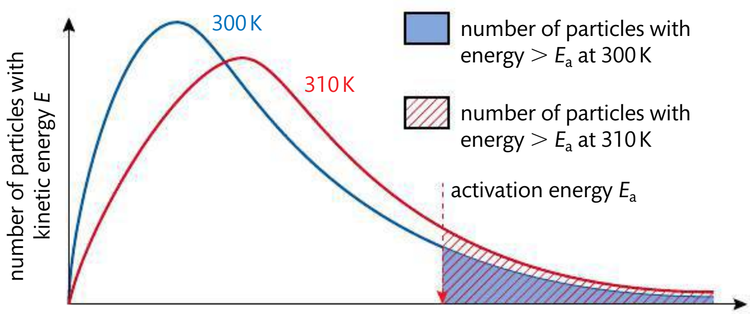
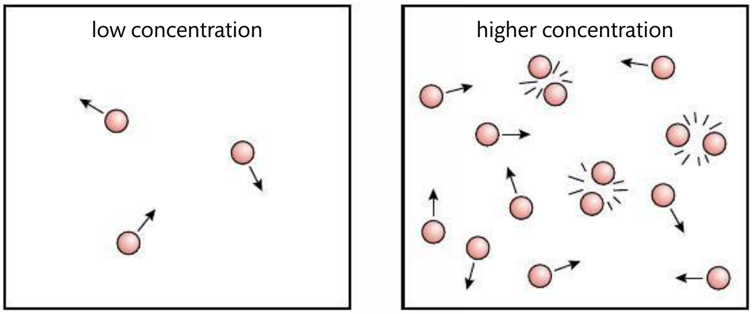


Figure 6.17 Maxwell–Boltzmann distribution curves for a sample of gas at two different temperatures, showing the higher proportion of particles with kinetic energy greater than the activation energy at the higher temperature.

So, as temperature increases, there is an increase in collision frequency due to the higher kinetic energy, but more importantly there is an increase in the frequency of collisions involving particles with the necessary activation energy to overcome the activation energy barrier. Consequently, there is an increase in the frequency of *successful* collisions and so an increase in the rate of reaction. Many reactions double their reaction rate for every 10 °C increase in temperature. In section 16.2 of this chapter, we will explore this further through a study of the mathematical relationship between temperature and rate of reaction.

2 Concentration

Increasing the concentration of reactants increases the rate of reaction. This is because as concentration increases, the frequency of collisions between reactant particles increases, as shown in Figure 6.18. The frequency of successful collisions therefore increases too.



When particles are closer together they have a greater chance of reacting.

Figure 6.18 Effect of concentration on collision frequency.

We can see the effect of concentration by following the rate of a reaction as it progresses. As reactants are used up, their concentration falls and the rate of the reaction decreases, giving the typical rate curve we saw in Figure 6.2.

The effect of concentration on the rate of reaction between zinc and sulfuric acid. The tube on the left has a more concentrated solution of the acid, the one on the right a more dilute solution. The product, hydrogen gas, is seen collecting much more quickly in the presence of the more concentrated acid.

3 Particle size

Decreasing the particle size increases the rate of reaction. This is important in heterogeneous reactions where the reactants are in different phases, such as a solid reacting with a solution. Subdividing a large particle into smaller parts increases the total surface area and therefore allows more contact and a higher probability of collisions between the reactants. You know, for example, how much easier it is to start a fire using small pieces of wood, rather than a large log – it's because with the small pieces there is more contact between the wood and the oxygen with which it is reacting. In reactions involving dispersions, stirring may help to decrease particle size and so increase the rate.

The effect of particle size can be demonstrated in the reaction between marble ($CaCO_3$) and hydrochloric acid. When marble chips are replaced with powder, the effervescence caused by carbon dioxide release is much more vigorous.

This effect of particle size on reaction rate can be quite dramatic. It has been responsible for many industrial accidents involving explosions of flammable dust powders – for example coal dust in mines and flour in mills.

Specimens of the extinct mammal the mammoth, dated as 10 000 years old, have been found perfectly preserved in the Arctic ice, whereas individual specimens of a similar age found in California, have only bones that remain. This is an illustration of the effect of the low temperature in the Arctic decreasing the rate of the reactions of decay. The same concept is used in the process of refrigerating or freezing food to preserve it.

4 Pressure

For reactions involving gases, increasing pressure increases the rate of reaction. This is because the higher pressure compresses the gas, effectively increasing its concentration. This will increase the frequency of collisions.

(a) Explain why they added water to the mixtures. (1)

(b) (i) Deduce the order of reaction for each substance and the rate expression from the results. (2)

(ii) Comment on whether Alex's or Hannah's hypothesis is correct. (1)

(c) Using the data from Experiment 1, determine the concentration of the substances used and the rate constant for the reaction including its units. (3)

(d) (i) This reaction uses a catalyst. Sketch and annotate the Maxwell–Boltzmann energy distribution curve for a reaction with and without a catalyst on labelled axes. (3)

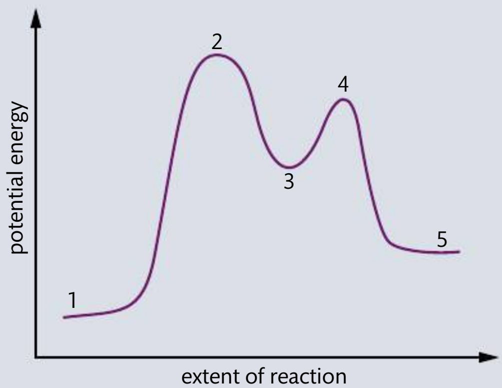
(ii) Describe how a catalyst works. (1)

(*Total 11 marks*)

15 The diagram below shows a potential energy level profile for a reaction. It occurs by the following mechanism:

$$W + XY \rightarrow WY + X$$

$$WY + Z \rightarrow W + YZ$$



(a) Deduce the equation for the overall reaction. (1)

(b) Label each step 1–5 as reactant, product, transition state, or intermediate. (3)

(c) Identify the species present at each step. (3)

(d) Determine which step is the rate-determining step and deduce the rate expression for the overall reaction. (2)

(e) Consider whether the reaction is catalysed, and if so identify the catalyst. (1)

(f) Add labelled arrows to the diagram to show the activation energy, E_a, and the enthalpy change, ΔH. (2)

(*Total 12 marks*)

To access weblinks on the topics covered in this chapter, please go to www.pearsonhotlinks.com and enter the ISBN or title of this book.

07 Equilibrium

Electron micrograph of a section through human lung tissue, showing air spaces called alveoli. Equilibrium considerations help us to understand how oxygen and carbon dioxide are exchanged between the blood and the air in these spaces.

Essential ideas

7.1 Many reactions are reversible. These reactions will reach a state of equilibrium when the rates of the forward reaction and reverse reaction are equal. The position of equilibrium can be controlled by changing the conditions.

17.1 The position of equilibrium can be quantified by the equilibrium law. The equilibrium constant for a particular reaction only depends on the temperature.

Imagine that you are part way along an escalator (a moving staircase) that is moving up and you decide to run down. If you can run down at exactly the same speed as the escalator is moving up, you will have no *net* movement. So, if someone were to take a picture of you at regular time intervals, it would seem as if you were not moving at all. Of course, in reality both you and the escalator *are* moving, but because there is no net change neither movement is observable. In chemical reactions a similar phenomenon occurs when a reaction takes place at the same rate as its reverse reaction, so no net change is observed. This is known as the **equilibrium state**.

In this chapter, we explore some of the features of the equilibrium state and learn how to derive and use the equilibrium constant expression. Application of the **equilibrium law** enables us to quantify reaction components and to predict how far reactions will proceed under different conditions. Industrial processes rely significantly on this type of study to determine the conditions that will maximize the yield of product. Equilibrium studies are also important in many biochemical and environmental processes, such as predicting the solubility of gases in the blood and knowing how certain chemicals in the atmosphere may react together to form pollutants that contribute to climate change. By the end of this chapter, you will be ready to tackle some of these applications in subsequent chapters.

Snow and ice-covered peak in the Cordillera Huayhuash, Peru. Equilibrium considerations help explain the relationship between water vapour in the clouds and precipitations in the form of liquid (rain) and solid (snow) at different temperatures and pressures.

7.1 Equilibrium

Understandings:

- A state of equilibrium is reached in a closed system when the rates of the forward and reverse reactions are equal.

  Guidance
  Physical and chemical systems should be covered.
- The equilibrium law describes how the equilibrium constant (K_c) can be determined for a particular chemical equation.
- The magnitude of the equilibrium constant indicates the extent of a reaction at equilibrium and is temperature dependent.
- The reaction quotient (Q) measures the relative amount of products and reactants present during a reaction at a particular point in time. Q is the equilibrium expression with non-equilibrium concentrations. The position of the equilibrium changes with changes in concentration, pressure, and temperature.
- A catalyst has no effect on the position of equilibrium or the equilibrium constant.

Applications and skills:

- The characteristics of chemical and physical systems in a state of equilibrium.
- Deduction of the equilibrium constant expression (K_c) from an equation for a homogeneous reaction.
- Determination of the relationship between different equilibrium constants (K_c) for the same reaction at the same temperature.

Guidance

Relationship between K_c values for reactions that are multiples or inverses of one another should be covered.

- Application of Le Chatelier's principle to predict the qualitative effects of changes of temperature, pressure, and concentration on the position of equilibrium and on the value of the equilibrium constant.

Guidance

Specific details of any industrial process are not required.

Bromine stored in a sealed jar. The system is in dynamic equilibrium, so the concentrations of liquid and vapour do not change at constant temperature.

Bromine is the only non-metallic element that is liquid at room temperature. It is extremely toxic and takes its name from a Greek word meaning stench.

Physical systems

Consider what happens when some bromine, Br_2, is placed in a sealed container at room temperature.

As bromine is a **volatile** liquid, with a boiling point close to room temperature, a significant number of particles (molecules of Br_2) will have enough energy to escape from the liquid state and form vapour in the process known as **evaporation**. As the container is sealed, the bromine vapour cannot escape and so its concentration will increase. Some of these vapour molecules will collide with the surface of the liquid, lose energy, and become liquid in the process known as **condensation**.

$$Br_2(l) \underset{\text{condensation}}{\overset{\text{evaporation}}{\rightleftharpoons}} Br_2(g)$$

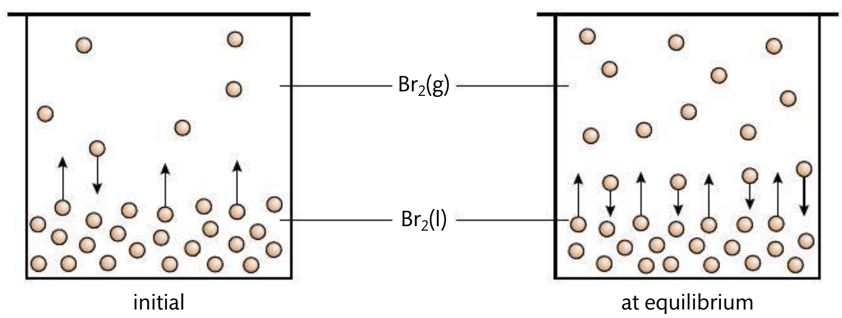


Figure 7.1 Establishing dynamic equilibrium in the evaporation of bromine. Equilibrium is established when the rate of evaporation equals the rate of condensation.

The rate of condensation increases with the increase in concentration of vapour, as more vapour particles collide with the surface of the liquid. Eventually, the rate of condensation is *equal* to the rate of evaporation, and at this point there will be no net change in the amounts of liquid and gas present (Figure 7.2). We say that the system has reached **equilibrium**. This will only occur in a closed system, where the $Br_2(g)$ cannot escape as vapour but may condense back into the liquid.

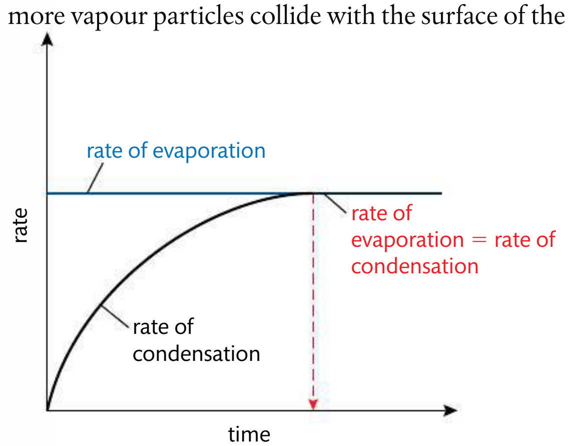


Figure 7.2 The rates of evaporation and condensation as liquid–vapour equilibrium is established in a closed system. The rate of evaporation is constant, while the rate of condensation increases with increasing concentration of vapour. Equilibrium is established when the two rates are equal.

Chemical systems

Consider the reaction of dissociation between hydrogen iodide (HI) and its elements hydrogen (H_2) and iodine (I_2). Hydrogen and hydrogen iodide are both colourless, whereas iodine is released as a purple gas, so this helps us to see what is happening.

$2HI(g)$	$\rightleftharpoons$	$H_2(g)$	+	$I_2(g)$
colourless gas		colourless gas		purple gas

If we carry out this reaction starting with hydrogen iodide in a sealed container, there will at first be an increase in the purple colour owing to the production of iodine gas. But after a while this increase in colour will stop, and it may appear that the reaction too has stopped. In fact, what has happened is that the rate of the dissociation of HI is fastest at the start when the concentration of HI is greatest and falls as the reaction proceeds. Meanwhile, the reverse reaction, which initially has a zero rate because there is no H_2 and I_2 present, starts slowly and increases in rate as the concentrations of H_2 and I_2 increase. Eventually, the rate of dissociation of HI has become equal to the rate of the reverse reaction of association between H_2 and I_2, so the concentrations remain constant. This is why the colour in the flask remains the same. At this point, equilibrium has been reached. It is described as **dynamic** because both forward and backward reactions are still occurring.

Iodine gas in a stoppered flask. Iodine is a crystalline solid at room temperature, but sublimes on heating to form a purple gas.

If we were to analyse the contents of the flask at this point, we would find that HI, H_2, and I_2 would all be present and that if there were no change in conditions, their concentrations would remain constant over time. We refer to this as the **equilibrium mixture**.

If we reversed the experiment and started with H_2 and I_2 instead of HI, we would find that eventually an equilibrium mixture would again be achieved in which the concentrations of H_2, I_2, and HI would remain constant. These relationships are shown in Figure 7.3.

TOK

The study of chemical change often involves both the macroscopic and microscopic scales. Which ways of knowing do we use in moving from the macroscopic to the microscopic?

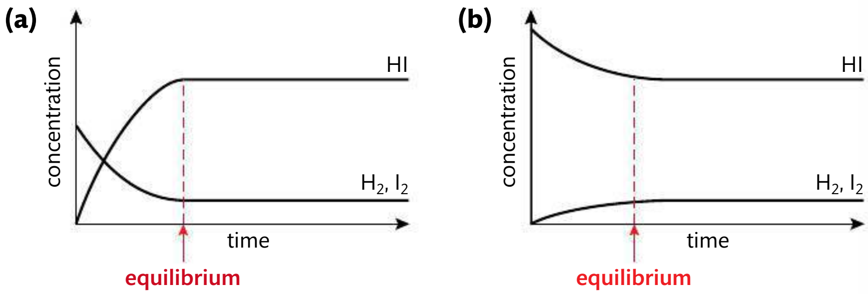


Figure 7.3 Equilibrium is reached when the concentrations of reactants and products become constant. Note the same equilibrium mixture is reached starting from (a) a mixture of H_2 and I_2 or (b) from pure HI.

The equilibrium state has specific characteristics

In studies of equilibria we are dealing with reversible reactions – those that occur in both directions. The convention is to describe the reaction from left to right (reactants to products) as the **forward reaction**, and the reaction from right to left (products to reactants) as the **backward** or **reverse reaction**. The symbol $\rightleftharpoons$ is used to show that the reaction is an equilibrium reaction.

$$\text{reactants} \underset{\text{backward reaction}}{\overset{\text{forward reaction}}{\rightleftharpoons}} \text{products}$$

The examples of physical and chemical systems discussed have shown that:

> " At equilibrium the rate of the forward reaction is equal to the rate of the backward reaction. "

Strictly speaking, all reactions can be considered as equilibrium reactions. However, in many cases the equilibrium mixture consists almost entirely of products – that is, it is considered to have gone virtually to completion. By convention we use the symbol $\rightarrow$ rather than the equilibrium symbol in these cases. In other reactions there may be so little product formed that it is undetectable and the reaction is considered to have effectively not happened.

Make sure that you use the equilibrium symbol $\rightleftharpoons$ when writing equations for reactions where the reverse reactions are significant. For example, it must be used when explaining the behaviour of weak acids and bases (Chapter 8).

At equilibrium the rate of the forward reaction is equal to the rate of the backward reaction.

What are the differences between theories and analogies as forms of explanation?

These reactions have also shown some of the main features of the equilibrium state, and these can now be summarized as they apply to *all* reactions at equilibrium.

	Feature of equilibrium state	Explanation
1	Equilibrium is dynamic	The reaction has not stopped but both forward and backward reactions are still occurring at the same rate.
2	Equilibrium is achieved in a closed system	A closed system has no exchange of matter with the surroundings, so equilibrium is achieved where both reactants and products can react and recombine with each other.
3	The concentrations of reactants and products remain constant at equilibrium	They are being produced and destroyed at an equal rate.
4	At equilibrium there is no change in macroscopic properties	Macroscopic properties are observable properties such as colour and density. These do not change as they depend on the concentrations of the components of the mixture.
5	Equilibrium can be reached from either direction	The same equilibrium mixture will result under the same conditions, no matter whether the reaction is started with all reactants, all products, or a mixture of both.

It is important to understand that even though the concentrations of reactant and product are *constant* at equilibrium, this in no way implies that they are *equal*. In fact, most commonly there will be a much higher concentration of either reactant or product in the equilibrium mixture, depending both on the reaction and on the conditions. We can see, for example, in Figure 7.3 that when the dissociation of HI reaches equilibrium, there is a higher concentration of HI than of H_2 and I_2.

Thinking back to the analogy in the introduction to this chapter, of you running in the opposite direction on a moving staircase where the top and bottom represent reactants and products respectively, it would be possible for you to be 'at equilibrium' near the top of the staircase, near the bottom, or anywhere in between. As long as you were moving at the same speed as the staircase you would still have no net change in position.

The proportion of reactant and product in the equilibrium mixture is referred to as its **equilibrium position**. Reactions where the mixture contains predominantly products are said to 'lie to the right' and reactions with predominantly reactants are said to 'lie to the left'. It is, however, often useful to be able to capture this information mathematically to compare the equilibrium mixtures of different reactions and the effect of different conditions. In the next section we will look at how this is done.

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At equilibrium no change is observed on the macroscopic level, although particles are reacting at the microscopic level. These changes can be deduced using techniques such as isotopic labelling, which allow the progress of a specific reactant to be followed. Our power of understanding is enhanced by contributions from instrumentation and sensors that may gather information beyond human sense perception.

Exercises

1 Which statements are correct for a reaction at equilibrium?

I The forward and reverse reactions both continue.
II The rates of the forward and reverse reactions are equal.
III The concentrations of reactants and products are equal.

A I and II only **B** I and III only **C** II and III only **D** I, II, and III

2 Which statement is always true for a chemical reaction that has reached equilibrium at constant temperature?

A The yield of product(s) is greater than 50%.
B The rate of the reverse reaction is lower than that of the forward reaction.
C The amounts of reactants and products do not change.
D Both forward and reverse reactions have stopped.

3 Which statement is *not* true for a mixture of ice and water at equilibrium at constant temperature?

A The rates of melting and freezing are equal.
B The amounts of ice and water are equal.
C The same position of equilibrium can be reached by cooling water and by heating ice.
D There is no observable change in the system.

Sometimes you may see the equilibrium sign written with unequal arrows such as ⇀̲↽. This is used to represent the reaction that lies in favour of products. Likewise, ↼̲⇁ is used to represent a reaction that lies in favour of reactants.

The equilibrium constant K_c can be predicted from a reaction's stoichiometry

Consider now the reaction

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

If we were to carry out a series of experiments on this reaction with different starting concentrations of H_2, I_2, and HI, we could wait until each reaction reached equilibrium and then measure the composition of each equilibrium mixture. Here are some typical results obtained at 440 °C.

Experiment I

	Initial concentration / mol dm^{-3}	Equilibrium concentration / mol dm^{-3}
H_2	0.100	0.0222
I_2	0.100	0.0222
HI	0.000	0.156

Experiment II

	Initial concentration / mol dm^{-3}	Equilibrium concentration / mol dm^{-3}
H_2	0.000	0.0350
I_2	0.0100	0.0450
HI	0.350	0.280

Experiment III

	Initial concentration / mol dm^{-3}	Equilibrium concentration / mol dm^{-3}
H_2	0.0150	0.0150
I_2	0.000	0.0135
HI	0.127	0.100

At a glance these data may not appear to show any pattern. However, there is a predictable relationship among the different compositions of these equilibrium mixtures, and the key to discovering it is in the stoichiometry of the reaction equation.

CHALLENGE YOURSELF

1 A closed system is defined differently in different disciplines. In thermodynamics it means that no matter can be exchanged with the surroundings, but energy can flow. To what extent can the Earth can be considered a closed system?

Remember square brackets [ ] are commonly used to show concentration in mol dm^{-3}.

Many sources give units for K_c which are a multiple of mol dm^{-3}, depending on the stoichiometry of the reaction. In fact this is not fully correct, as the terms in the equilibrium expression are really a thermodynamic quality known as 'activity' that has no units. For this reason, we are omitting them in the values of K_c here, and you will not be required to include them in IB examination answers.

The equilibrium constant K_c has a fixed value for a particular reaction at a specified temperature. The only thing that changes the value of K_c for a reaction is the temperature.

The equilibrium constant expression will only give the value K_c when the concentrations used in the equation are the *equilibrium* concentrations for all reactants and products. Strictly speaking, the subscript 'eqm' should always be used in the equation, but by convention this is generally left out. However, make completely sure that the only values you substitute into an equation to calculate K_c are the equilibrium concentrations.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

If we take the *equilibrium* concentrations and process them in the following way:

$$\frac{[HI]^2_{eqm}}{[H_2]^1_{eqm}[I_2]^1_{eqm}}$$

2 = coefficient of HI in the reaction equation

1 = coefficient of H_2 in the reaction equation

1 = coefficient of I_2 in the reaction equation

we find the following results:

Experiment I	Experiment II	Experiment III
$\frac{(0.156)^2}{0.0222\times0.0222}=49.4$	$\frac{(0.280)^2}{0.035\times0.0450}=49.8$	$\frac{(0.100)^2}{0.0150\times0.0135}=49.4$

Clearly this way of processing the equilibrium data produces a constant value within the limits of experimental accuracy. This constant is known as the **equilibrium constant**, $\boldsymbol{K_c}$. It has a fixed value for this reaction *at a specified temperature*.

In fact, every reaction has its own particular value of K_c which can be derived in a similar way. First we use the balanced reaction equation to write the **equilibrium constant expression**.

For the reaction:

$$aA + bB \rightleftharpoons cC + dD$$

the equilibrium constant expression is

$$\frac{[C]^c_{eqm}[D]^d_{eqm}}{[A]^a_{eqm}[B]^b_{eqm}} = K_c$$

The value for K_c can then be determined by substituting the equilibrium concentrations of all reactants and products into this equation.

Note:

- The equilibrium constant expression has the concentrations of products in the numerator and the concentrations of reactants in the denominator.
- Each concentration is raised to the power of its coefficient in the balanced equation. (Where it is equal to one it does not have to be given.)
- Where there is more than one reactant or product the terms are multiplied together.

Worked example

Write the equilibrium expression for the following reactions.

(i) $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$

(ii) $Cu^{2+}(aq) + 4NH_3(aq) \rightleftharpoons [Cu(NH_3)_4]^{2+}$

Solution

(i) $K_c = \frac{[H_2O]^2}{[H_2]^2[O_2]}$

(ii) $K_c = \frac{[[Cu(NH_3)_4]^{2+}]}{[Cu^{2+}][NH_3]^4}$

Exercises

4 Write the equilibrium constant expression for the following reactions:

(a) $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$
(b) $4NH_3(g) + 7O_2(g) \rightleftharpoons 4NO_2(g) + 6H_2O(g)$
(c) $CH_3Cl(aq) + OH^-(aq) \rightleftharpoons CH_3OH(aq) + Cl^-(aq)$

5 Write the equations for the reactions represented by the following equilibrium constant expressions

(a) $K_c = \dfrac{[NO_2]^2}{[N_2O_4]}$

(b) $K_c = \dfrac{[CO][H_2]^3}{[CH_4][H_2O]}$

6 Write the equilibrium constant expressions for the following chemical reactions:

(a) fluorine gas and chlorine gas combine to form $ClF_3(g)$
(b) NO dissociates into its elements
(c) methane, CH_4, and steam react to form carbon monoxide and hydrogen

The equilibrium constant expressions described here apply to homogeneous reactions, that is reactions where reactants and products are in the same phase, as gases, liquids, or in solution. It is good practice always to include state symbols in your equations to ensure this is being applied correctly.

The magnitude of K_c gives information on the extent of reaction

Different reactions have different values of K_c. What does this value tell us about a particular reaction?

As the equilibrium constant expression puts products on the numerator and reactants on the denominator, a high value of K_c will mean that at equilibrium there are proportionately more products than reactants. In other words, such an equilibrium mixture lies to the right and the reaction goes almost to completion. By contrast, a low value of K_c must mean that there are proportionately less products with respect to reactants, so the equilibrium mixture lies to the left and the reaction has barely taken place.

Consider the following three reactions and their K_c values measured at 550 K.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ $\quad K_c = 2$

$H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$ $\quad K_c = 10^{10}$

$H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$ $\quad K_c = 10^{18}$

The large range in their K_c values tells us about the differing extents of these reactions. We can deduce that the reaction between H_2 and Cl_2 has taken place the most fully at this temperature, while H_2 and I_2 have reacted the least.

A good rule of thumb to apply to these situations is that if $K_c \gg 1$, the reaction is considered to go almost to completion (very high conversion of reactants into products), and if $K_c \ll 1$, the reaction hardly proceeds.

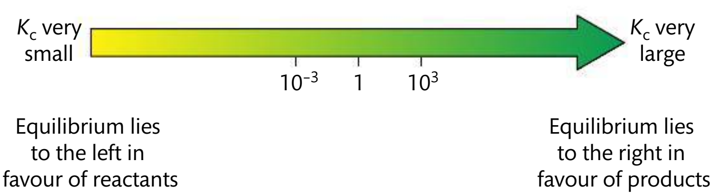

Figure 7.4 The larger the value of K_c, the further the equilibrium mixture lies to the right.

Note that the magnitude of K_c does not give us any information on the *rate* of the reaction. It informs us of the nature of the equilibrium mixture, but not on how quickly the equilibrium state will be achieved.

CHALLENGE YOURSELF

2 Why do you think the reactions of the three halogens Cl_2, Br_2, and I_2, with H_2 have such different values for their equilibrium constant at the same temperature? What can you conclude about the strength of bonding in the three hydrogen halides?

The magnitude of the equilibrium constant K_c gives information about how far a reaction goes at a particular temperature, but not about how fast it will achieve the equilibrium state.

The reaction quotient, Q, enables us to predict the direction of reaction

Remember that the value of K_c is calculated from substituting the *equilibrium* concentrations of all reactants and products into the equilibrium constant expression. Any other values used will not give the equilibrium value K_c for that reaction at that temperature.

This in itself can be useful. If we take the concentrations of the reactants and products at one moment in time when the reaction is not at equilibrium, and substitute these into the equilibrium constant expression, we obtain a value known as the **reaction quotient**, **Q**. As time passes and the reaction continues, the concentrations of all reaction components change and eventually reach the equilibrium concentrations. In other words, the value of Q changes in the direction of K_c. This enables us to predict the direction in which the reaction will proceed.

For example, if we again consider the reaction:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

for which $K_c = 49.5$ at 440 °C, as we saw on page 316.

The table below shows experimental data for the concentrations of the reaction components from experiments I and II at a time when the reaction mixture was not at equilibrium.

	Experiment I: concentration at time t / mol dm^{-3}	Experiment II: concentration at time t / mol dm^{-3}
H_2	0.0500	0.0250
I_2	0.0500	0.0350
HI	0.100	0.300

The equilibrium constant expression = $\dfrac{[HI]^2}{[H_2][I_2]}$

Experiment I, time t $\quad Q = \dfrac{(0.100)^2}{(0.0500)(0.0500)} = 4.00$

Experiment II, time t $\quad Q = \dfrac{(0.300)^2}{(0.0250)(0.0350)} = 103$

In experiment I, $Q < K_c$ and so Q must increase as the reaction moves towards equilibrium. This means that the net reaction must be to the right, in favour of products.

In experiment II, $Q > K_c$ and so Q must decrease as the reaction moves towards equilibrium. This means that the net reaction must be to the left, in favour of reactants.

We can summarize the use of the reaction quotient, Q, in determining the direction of reaction as follows:

- if $Q = K_c$, reaction is at equilibrium, no net reaction occurs;
- if $Q < K_c$, reaction proceeds to the right in favour of products;
- if $Q > K_c$, reaction proceeds to the left in favour of reactants.

The reaction quotient, Q, is a measure of the relative amounts of reactants and products present in a reaction at a particular time. Its value is determined from substituting concentrations of reaction components, all measured at the same time, into the equilibrium expression.

Note that the value of Q for a reaction does not have a fixed value as it can be measured at any point in time, whereas the value of K_c for a reaction is a constant at a specified temperature.

Worked example

The equilibrium constant K_c for the reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

is 1.7×10^2 at 500 K.

Determine whether the reaction mixture is at equilibrium when the concentrations of the components at this temperature are as follows:

$[N_2] = 1.50$

$[H_2] = 1.00$

$[NH_3] = 8.00$.

If it is not at equilibrium, state and explain in which direction the reaction will proceed.

Solution

First write the equilibrium constant expression.

$$\frac{[NH_3]^2}{[N_2][H_2]^3}$$

Calculate the value of Q at these conditions by substituting the given concentration values into the equilibrium expression.

$$Q = \frac{(8.00)^2}{(1.50)(1.00)^3} = 42.7$$

Compare the value of Q at the given conditions with K_c.

$$42.7 < 1.7 \times 10^2$$

Therefore the reaction is not at equilibrium. It will proceed to the right in favour of products, as the value of Q must increase to be equal to K_c at equilibrium.

Comparison of the value of the reaction quotient, *Q*, at a specific time in a reaction with the value of K_c for the reaction, enables us to predict the direction of change.

The reaction quotient Q is used in the calculation of the electrode potential of a half-cell operating under non-standard conditions in the Nernst equation. This is discussed in topic C.6 in Chapter 14, Energy.

Relationships between K_c for different equations of a reaction

As K_c is defined with products on the numerator and reactants on the denominator, each raised to the power of their stoichiometric coefficients in the balanced equation, we can manipulate its value according to changes made to these terms. For this discussion we will consider the generic reaction:

$$aA + bB \rightleftharpoons cC + dD$$

for which the equilibrium constant is:

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

CHALLENGE YOURSELF

3 In heterogeneous equilibrium, the terms for pure solids or liquids are not shown in the equilibrium expression. Suggest why this is the case.

1 K_c for the inverse reaction

The inverse reaction

$$cC + dD \rightleftharpoons aA + bB$$

defines the products as reactants and vice versa. We will denote its equilibrium constant as K_c'.

$$K_c' = \frac{[A]^a[B]^b}{[C]^c[D]^d}$$

We can see that $K_c' = \frac{1}{K_c}$ or $K_c' = K_c^{-1}$.

In other words the equilibrium constant for a reaction is the reciprocal of the equilibrium constant for its inverse reaction.

2 K_c for a multiple of a reaction

Consider now the reaction

$$2aA + 2bB \rightleftharpoons 2cC + 2dD$$

We will denote its equilibrium constant as K_c^x.

$$K_c^x = \frac{[C]^{2c}[D]^{2d}}{[A]^{2a}[B]^{2b}}$$

We can see that each term has been squared in K_c^x relative to its value in K_c.

So $K_c^x = K_c^2$.

By similar thinking, we can conclude that tripling of the stoichiometric coefficients would lead to a cubing of the value of K_c, halving of the stoichiometric coefficients would lead to a square rooting of the value of K_c, and so on.

These manipulations of the value of K_c are summarized below.

	Effect on equilibrium expression	Effect on K_c
inversing the reaction	inverts the expression	$\frac{1}{K_c}$ or K_c^{-1}
doubling the reaction coefficients	squares the expression	K_c^2
tripling the reaction coefficients	cubes the expression	K_c^3
halving the reaction coefficients	square roots the expression	$\sqrt{K_c}$
adding together two reactions	multiplies the two expressions	$K_c^{i} \times K_c^{ii}$

Worked example

The equilibrium constant for the reaction

$$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$$

is 0.04 at a certain temperature. What would be the value of the equilibrium constant, K_c', for the following reaction at the same temperature?

$$\tfrac{1}{2}H_2(g) + \tfrac{1}{2}I_2(g) \rightleftharpoons HI(g)$$

Solution

$$K_c = \frac{[H_2][I_2]}{[HI]^2}$$

$$K_c' = \frac{[HI]}{[H_2]^{1/2}[I_2]^{1/2}}$$

Note that the reaction is reversed and the coefficients in the equation are halved. Overall, the value of K_c' is the square root of the value of the reciprocal of K_c.

So $K_c' = \dfrac{1}{\sqrt{K_c}}$ or $\sqrt{(K_c)^{-1}}$.

$\therefore K_c' = \sqrt{\dfrac{1}{0.04}} = 5.0$

Exercises

7 When the following reactions reach equilibrium, does the equilibrium mixture contain mostly reactants or mostly products at the specified temperature? Assume that the value for K_c given corresponds to the temperature of the reaction mixture.

(a) $N_2(g) + 2H_2(g) \rightleftharpoons N_2H_4(g)$ $K_c = 7.4 \times 10^{-26}$
(b) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ $K_c = 2.7 \times 10^{-18}$
(c) $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ $K_c = 6.0 \times 10^{13}$

8 The equilibrium constant for the reaction:

$$H_2O(g) + Cl_2O(g) \rightleftharpoons 2HOCl(g)$$

is 0.0900 at 298 K.

Determine whether the following sets of conditions represent an equilibrium mixture for the reaction at this temperature. For those not at equilibrium, determine in which direction the reaction will proceed.

	$[H_2O]$	$[Cl_2O]$	$[HOCl]$
(a)	0.100	0.100	1.00
(b)	0.49	0.040	0.042
(c)	0.19	0.00033	0.083

9 At a given temperature, the reaction

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

has a value of $K_c = 278$. Determine values of K_c for the following reactions at this temperature.

(a) $4SO_2(g) + 2O_2(g) \rightleftharpoons 4SO_3(g)$
(b) $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$
(c) $SO_3(g) \rightleftharpoons SO_2(g) + \tfrac{1}{2}O_2(g)$

When equilibrium is disrupted

A system remains at equilibrium so long as the rate of the forward reaction equals the rate of the backward reaction. But as soon as this balance is disrupted by any change

Henri-Louis Le Chatelier was a French chemist who published his equilibria principle in 1884. Amongst other research, he also investigated the possibility of ammonia synthesis, but abandoned his efforts after suffering a devastating explosion in his laboratory. After Haber's later elucidation of the conditions required in the reaction, Le Chatelier realized that he had been very close to the discovery himself. Late in his life he wrote 'I let the discovery of the ammonia synthesis slip through my hands. It was the greatest blunder of my career.' We can only speculate on how history might have been re-written if this discovery had in fact been made in France rather than in Germany before World War I.

When a system at equilibrium is subjected to a change, it will respond in such a way as to minimize the effect of the change.

Experiment to show the effect of changing the concentration of chloride ions on the cobalt chloride equilibrium reaction:

$[Co(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons CoCl_4^{2-}(aq) + 6H_2O(l)$

The flask on the left has a low concentration of chloride ions, giving the pink colour of the complex ion with water. As the concentration of chloride ions is increased, the equilibrium shifts to the right, changing the colour from pink to blue. Adding water would shift the equilibrium in the opposite direction. Cobalt chloride is often used to test for the presence of water because of this colour change.

in conditions that unequally affects the rates of these reactions, the equilibrium condition will no longer be met. It has been shown, however, that equilibria respond in a predictable way to such a situation, based on a principle known as **Le Chatelier's principle**. This states that *a system at equilibrium when subjected to a change will respond in such a way as to minimize the effect of the change*. Simply put, this means that whatever we do to a system at equilibrium, the system will respond in the opposite way. Add something and the system will react to remove it, remove something and the system will react to replace it. After a while, a new equilibrium will be established and this will have a different composition from the earlier equilibrium mixture. Applying the principle therefore enables us to predict the qualitative effect of typical changes that occur to systems at equilibrium.

Changes in concentration

Suppose an equilibrium is disrupted by an increase in the concentration of one of the reactants. This will cause the rate of the forward reaction to increase while the backward reaction will not be affected, so the reaction rates will no longer be equal. When equilibrium re-establishes itself, the mixture will have new concentrations of all reactants and products, and the equilibrium will have shifted in favour of products. The value of K_c will be unchanged. This is in keeping with the prediction from Le Chatelier's principle: addition of reactant causes the system to respond by removing reactant – this favours the forward reaction and so shifts the equilibrium to the right.

Similarly, the equilibrium could be disrupted by a decrease in the concentration of product by removing it from the equilibrium mixture. As the rate of the backward reaction is now decreased, there will be a shift in the equilibrium in favour of the products. A different equilibrium position will be achieved, but the value of K_c will be unchanged. Again this confirms the prediction from Le Chatelier's principle: removal of product causes the system to respond by making more product – this favours the forward reaction and so shifts the equilibrium to the right.

The graph in Figure 7.5 illustrates these disruptions to equilibrium for the reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

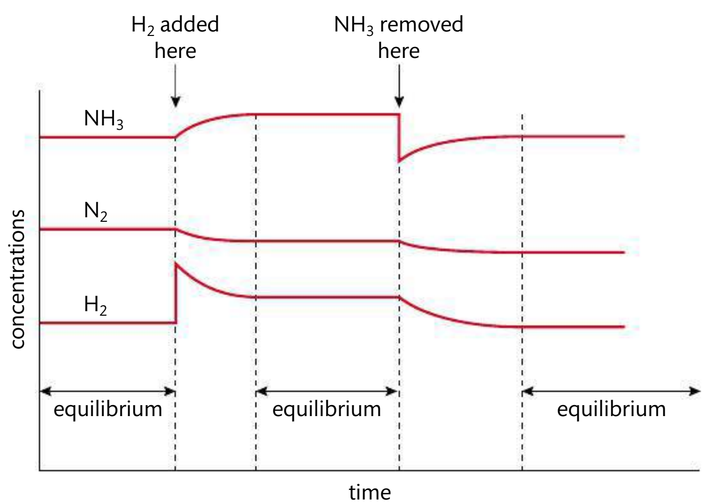

Figure 7.5 Effects of the addition of reactant and removal of product on the equilibrium $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$. When H_2 is added some N_2 reacts and more NH_3 is formed as the equilibrium shifts to the right. When NH_3 is removed, more N_2 reacts with H_2 as the equilibrium again shifts to the right. After each change a new equilibrium mixture is achieved.

After H_2 is added, the concentrations of N_2 and H_2 decrease (in a 1 : 3 ratio in keeping with their reaction stoichiometry), while the concentration of NH_3 rises (in a 2 : 1 ratio relative to N_2) as the rate of the forward reaction increases and the equilibrium shifts to the right. The new equilibrium mixture has a higher proportion of products. In the second part of the graph, removal of NH_3 from the equilibrium mixture causes a decrease in the concentrations of N_2 and H_2 as they react to form more NH_3.

Often in an industrial process the product will be removed as it forms. This ensures that the equilibrium is continuously pulled to the right, so increasing the yield of product.

Applying Le Chatelier's principle, can you think what concentration changes would cause an equilibrium to shift to the left? The answer is either an increase in concentration of product or a decrease in concentration of reactant.

Changes in pressure

Equilibria involving gases will be affected by a change in pressure if the reaction involves a change in the number of molecules. This is because there is a direct relationship between the number of gas molecules and the pressure exerted by a gas in a fixed volume. So if such a reaction at equilibrium is subject to an increase in pressure, the system responds to decrease this pressure by favouring the side with the smaller number of molecules. Conversely, a decrease in pressure will cause a shift in the equilibrium position to the side with the larger number of molecules of gas. A different equilibrium position will be achieved but the value of K_c will be unchanged, so long as the temperature remains the same.

For example, consider the reaction used in the production of methanol:

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

In total there are three molecules of gas on the left side and one molecule of gas on the right side. So here high pressure will shift the equilibrium to the right, in favour of the smaller number of gas molecules, which increases the yield of CH_3OH.

Note that many common equilibrium reactions do not involve a change in the number of gas molecules and so are not affected by changes in pressure. For example, the reaction

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

has two molecules of gas on both sides of the equation. Changing pressure for this reaction will affect the rate of the reaction but not the position of equilibrium or the value of K_c.

Changes in temperature

We have noted that K_c is temperature dependent, so changing the temperature will change K_c. However, in order to predict *how* it will change we must examine the enthalpy changes (see Chapter 5) of the forward and backward reactions. Remember that an exothermic reaction releases energy (ΔH negative), whereas an endothermic reaction absorbs energy (ΔH positive). The enthalpy changes of the forward and backward reactions are equal and opposite to each other.

So if we apply Le Chatelier's principle, including the energy change in the chemical reaction, we can predict how the reaction will respond to a change in temperature.

Consider the reaction:

$$\underset{\text{brown}}{2NO_2(g)} \rightleftharpoons \underset{\text{colourless}}{N_2O_4(g)} \qquad \Delta H = -57\ \text{kJ mol}^{-1}$$

Changes in the concentration of reactants or products alter the equilibrium position and so change the composition of the equilibrium mixture. But the value of K_c stays the same.

Changes in pressure or volume will affect the position of equilibrium of a reaction if it involves a change in the number of gas molecules. The value of K_c remains unchanged.

When describing the effects of volume or pressure on equilibrium reactions, you must state it depends on the relative number of *gas* molecules on both sides of the equation. Solids and liquids are hardly affected by changes in pressure.

An increase in pressure favours the side of an equilibrium reaction that has the smaller number of gas molecules.

When ΔH is given for an equilibrium reaction, by convention its sign refers to the forward reaction. So a negative sign for ΔH means that the forward reaction is exothermic, and the backward reaction is endothermic.

CHALLENGE YOURSELF

4 Use information from this section to explain why there is very little NO in the atmosphere under ordinary conditions, and why severe air pollution is often characterized by a brownish haze.

Experiment to show the effect of temperature on the reaction that converts NO_2 (brown) to N_2O_4 (colourless). As the temperature is increased, more NO_2 is produced and the gas becomes darker, as seen in the tube on the left.

The negative sign of ΔH tells us that the forward reaction is exothermic and so releases heat. If this reaction at equilibrium is subjected to a decrease in temperature, the system will respond by producing heat and it does this by favouring the forward exothermic reaction. This means that the equilibrium will shift to the right, in favour of the product N_2O_4. A new equilibrium mixture will be achieved and the value of K_c will increase. So here we can see that the reaction will give a higher yield of products at a *lower temperature*.

Conversely, increasing the temperature favours the backward endothermic reaction, and so shifts the equilibrium to the left, decreasing the value of K_c. The reaction mixture becomes a darker colour as the concentration of NO_2 increases.

The table below illustrates the effect of temperature on the value of K_c for this reaction.

Temperature / K	K_c for $2NO_2(g) \rightleftharpoons N_2O_4(g)$
273	1300
298	170

Even though in this case a lower temperature will produce an equilibrium mixture with a higher proportion of products, remember from Chapter 6 that low temperature also causes a lower rate of reaction. And so, although a higher yield will be produced eventually, it may simply take too long to achieve this from practical and economic considerations if this was an industrial-scale reaction. We will come back to this point later in this chapter.

Now consider the following reaction:

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g) \qquad \Delta H = +181\ \text{kJ mol}^{-1}$$

In this case we can see that the forward reaction is endothermic and so absorbs heat. So here the effect of a decreased temperature will be to favour the backward exothermic reaction. Therefore the equilibrium will shift to the left, in favour of reactants, and K_c will decrease. At higher temperatures, the forward endothermic reaction is favoured, so the equilibrium shifts to the right and K_c will increase.

The table below illustrates the effect of temperature on the value of K_c for this reaction.

Temperature / K	K_c for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
298	4.5×10^{-31}
900	6.7×10^{-10}
2300	1.7×10^{-3}

TOK

Combustion reactions are generally described as exothermic, but the reaction

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$$

is endothermic. When is it appropriate to refer to an 'exception to a rule', and when does the rule need to be reconsidered?

The reaction

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$$

takes place in motor vehicles where the heat released by the combustion of the fuel is sufficient to cause the nitrogen and oxygen gases from the air to combine together in this way. Unfortunately, the product NO is toxic, and, worse still, quickly becomes converted into other toxins that form the components of acid rain and smog. It is therefore of great interest to vehicle manufacturers to find ways of lowering the temperature during combustion in order to reduce the production of NO in the reaction above.

Increasing the temperature causes an increase in the value of K_c for an endothermic reaction and a decrease in the value of K_c for an exothermic reaction.

These examples illustrate that, unlike changes in concentration and pressure, changes in temperature *do* cause the value of K_c to change. An increase in temperature increases the value of K_c for an endothermic reaction and decreases the value of K_c for an exothermic reaction. This is because changes in temperature have a different effect on the rates of the forward and backward reactions, due to their different activation energies, as discussed in Chapter 6. In the next chapter, we will use this fact to explain why the pH of pure water is temperature dependent.

Addition of a catalyst

As we learned in Chapter 6, a catalyst speeds up the rate of a reaction by providing an alternate reaction pathway that has a transition state with a lower activation energy, E_a. This increases the number of particles that have sufficient energy to react without raising the temperature.

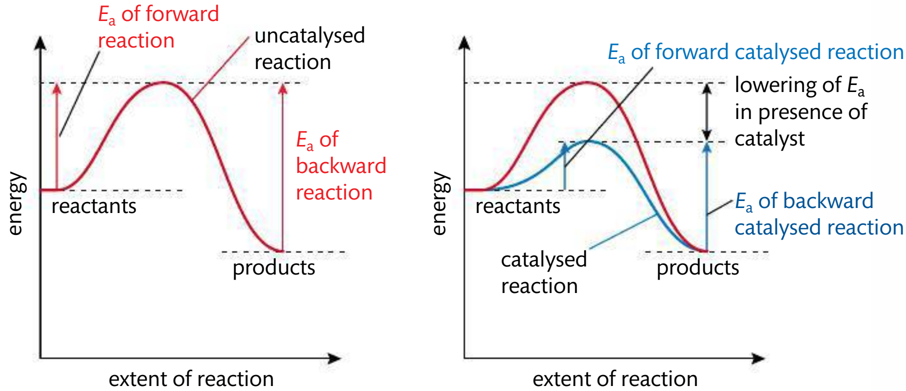

Figure 7.6 The effect of a catalyst in lowering the activation energy of both forward and backward reactions.

Because the forward and backward reactions pass through the same transition state, a catalyst lowers the activation energy by the same amount for the forward and backward reactions. So the rate of both these reactions will be increased by the same factor, as shown in Figure 7.6. The catalyst will therefore have no effect on the position of equilibrium, or on the value of K_c. In other words the catalyst will not increase the equilibrium yield of product in a reaction. It will, however, speed up the attainment of the equilibrium state and so cause products to form more quickly. Catalysts are generally not shown in the reaction equation or in the equilibrium constant expression as they are not chemically changed at the end of the reaction, and have no effect on the equilibrium concentrations.

Observations of shifts in the position of chemical equilibria
Full details of how to carry out this experiment with a worksheet are available online.

Catalysts are widely used in industrial processes to increase the rate of product formation, and are involved in every single biochemical reaction. One of the key principles of Green Chemistry is that catalytic reagents, chosen to be as selective as possible, are superior to stoichiometric reagents as they are not consumed in the reaction. Catalysts, including the differences between homogeneous and heterogeneous catalysts, are discussed in more detail in Option A Materials, topic A.3.

Catalysts do not change the position of equilibrium or the yield of a single reaction, but they enable the equilibrium mixture to be achieved more quickly. Catalysts do not change the value of K_c.

Summary

We can now summarize the effects of concentration, pressure, temperature, and catalyst on the position of equilibrium and on the value of K_c.

Effect of …	Change in position of equilibrium	Change in value of K_c
1 concentration	changes	no change
2 pressure	changes if reaction involves a change in the number of gas molecules	no change
3 temperature	changes	changes
4 catalyst	no change	no change

A particular reaction at a specified temperature can have many different possible equilibrium positions, but only one value for the equilibrium constant K_c.

NATURE OF SCIENCE

Knowledge of how systems at equilibrium respond to change has many applications, as we will see in the next section. Le Chatelier's principle is a useful predictive tool for this purpose, and has the advantage of being relatively simple to apply and effective. But it has some disadvantages too. Firstly, it leads only to qualitative predictions of the system's response. In some cases, such as investigations of when a precipitate will form in aqueous solution, this is insufficient. Secondly, Le Chatelier's principle offers no *explanation* for the response – it predicts what will happen, but not why. It can even be misleading in that it can suggest a false sense of purpose to systems' responses if the language used is such as 'the system tries to...'.

On the other hand, use of the reaction quotient, Q, to predict the consequence of change can often provide an effective explanation. As disturbances in concentration or pressure change the value of Q, the system responds as Q progresses towards K_c. But changes in temperature are different because they change the value of K_c – in this case the system responds as Q progresses towards the new value of K_c. The response of K_c to temperature can be explained through thermodynamics and the fact that the forward and backward reactions respond differently to temperature changes due to their different activation energies. Increased temperature favours the endothermic reaction as it has the higher activation energy. Scientific explanations often draw on different approaches and theories in this way. So long as the approaches do not contradict each other, together they can clarify and strengthen the understanding.

Note that you are not expected to learn specific conditions for any reaction, so there is no need to focus on the names of catalysts, specific temperature used, etc. But you should be able to apply an understanding of equilibria to any given example, and predict the conditions likely to be effective.

Exercises

10 The manufacture of sulfur trioxide can be represented by the equation below:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H^{\ominus} = -197\ \text{kJ mol}^{-1}$$

What happens when a catalyst is added to an equilibrium mixture from this reaction?

A The rate of the forward reaction increases and that of the reverse reaction decreases.
B The rates of both forward and reverse reactions increase.
C The value of $\Delta H^{\ominus}$ increases.
D The yield of sulfur trioxide increases.

11 What will happen to the position of equilibrium and the value of the equilibrium constant when the temperature is increased in the following reaction?

$$Br_2(g) + Cl_2(g) \rightleftharpoons 2BrCl(g) \qquad \Delta H^{\ominus} = +14\ \text{kJ}$$

	Position of equilibrium	Value of equilibrium constant
A	shifts towards the reactants	decreases
B	shifts towards the reactants	increases
C	shifts towards the products	decreases
D	shifts towards the products	increases

12 Which changes will shift the position of equilibrium to the right in the following reaction?

$$2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g)$$

I adding a catalyst
II decreasing the oxygen concentration
III increasing the volume of the container

A I and II only **B** I and III only **C** II and III only **D** I, II, and III

13 For each of the following reactions, predict in which direction the equilibrium will shift in response to an increase in pressure:

(a) $2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g)$
(b) $CO(g) + 2H_2 \rightleftharpoons CH_3OH(g)$
(c) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$

14 How will the equilibrium:

$$CH_4(g) + 2H_2S(g) \rightleftharpoons CS_2(g) + 4H_2(g) \qquad \Delta H = +ve$$

respond to the following changes?

(a) addition of $H_2(g)$
(b) addition of $CH_4(g)$
(c) a decrease in the volume of the container
(d) removal of $CS_2(g)$
(e) increase in temperature

Fritz Haber 1868–1934

15 The reaction

$$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g) \qquad \Delta H = -566 \text{ kJ mol}^{-1}$$

takes place in catalytic converters in cars. If this reaction is at equilibrium, will the amount of CO increase, decrease, or stay the same when:

(a) the pressure is increased by decreasing the volume?
(b) the pressure is increased by adding $O_2(g)$?
(c) the temperature is increased?
(d) a platinum catalyst is added?

Equilibrium theory is applied in many industrial processes

In reactions involving the manufacture of a chemical it is obviously a goal to obtain as high a conversion of reactant to product as possible. Application of Le Chatelier's principle enables chemists to choose conditions that will cause the equilibrium to lie to the right, and so help to achieve this. But the *yield* of a reaction is only part of the consideration. The *rate* is also clearly of great significance as it would be of limited value if a process were able to claim a high equilibrium yield of product, but took several years to achieve this. The economics of the process will depend on considerations of both the equilibrium and the kinetics of the reaction – in other words on how far and how fast the reaction will proceed. Sometimes these two criteria work against each other, and so the best compromise must be reached. A few case studies of industrial processes are discussed here.

The Haber process: the production of ammonia, NH_3

Fritz Haber was born in what is now Poland but moved to Germany early in his career. Together with Carl Bosch, also of Germany, he developed the process for the industrial synthesis of ammonia from its elements, and the first factory for ammonia production opened in Germany in 1913, just before World War I. This development had enormous significance for the country at war: it enabled the continued production of explosives despite the fact that imports of nitrate from Chile, used for producing nitric acid and explosives such as TNT, were barred through the blockaded ports. This effectively enabled Germany to continue its war efforts for another 4 years. Haber was awarded the Nobel Prize in Chemistry in 1918. In many ways history has recorded this as a controversial choice – not only had Haber's work helped to prolong the war, he had also been responsible for the development and use of chlorine as the first poison gas. Ironically, despite his evident patriotism towards Germany, he was expelled from the country in 1933 when the rising tide of anti-Semitism conflicted with his Jewish ancestry.

The book *The alchemy of air: a Jewish genius, a doomed tycoon, and the scientific discovery that fed the world but fueled the rise of Hitler* is written by Thomas Hager. It is a fascinating account of the historic discovery of the process to synthesize ammonia, and its continuing mixed consequences – these include the revolution in global food production, the death of millions through wars, and growing concerns of nitrate pollution and obesity.

The Haber process is based on the reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -93 \text{ kJ mol}^{-1}$$

The following information can be derived from this equation:

- all reactants and products are gases;
- there is a change in the number of gas molecules as the reaction proceeds: four gas molecules on the left and two on the right;

It is estimated that as much as 130 million tonnes of ammonia, NH_3, are produced worldwide every year. China is responsible for nearly one-third of this, and India, Russia, and the USA also produce significant amounts. Approximately 80% of the ammonia is used to make fertilizers, notably ammonium nitrate, NH_4NO_3. Other uses include synthesis of textiles such as nylon and powerful explosives. In April 2013, a major explosion occurred in a fertilizer factory in Texas, USA, killing and injuring hundreds of people. It was caused by a fire in a stock of 250 000 kilograms of ammonium nitrate. Accidents such as this in industrial plants raise many questions regarding health and safety, storage, and handling of chemicals and the balance of responsibility between companies, governments, and individuals.

Tractor applying a chemical solution of fertilizer to the soil. Ammonium salts such as ammonium nitrate and sulfate are particularly effective fertilizers as they supply nitrogen needed by plants and are readily soluble. It is estimated that without the use of ammonium fertilizers, two billion people would starve.

- the forward reaction is exothermic so releases heat; the backward reaction is endothermic so absorbs heat.

Application of Le Chatelier's principle to this reaction enables us to consider the optimum conditions.

- Concentration: the reactants nitrogen and hydrogen are supplied in the molar ratio 1 : 3 in accordance with their stoichiometry in the equation. The product ammonia is removed as it forms, thus helping to pull the equilibrium to the right and increasing the yield.
- Pressure: as the forward reaction involves a decrease in the number of gas molecules, it will be favoured by a *high pressure*. The usual pressure used in the Haber process is about 2×10^7 Pa.
- Temperature: as the forward reaction is exothermic, it will be favoured by a lower temperature. However, too low a temperature would cause the reaction to be uneconomically slow, and so a *moderate temperature* of about 450 °C is used.
- Catalyst: a catalyst will speed up the rate of production and so help to compensate for the moderate temperature used. A catalyst of finely divided iron is used, with small amounts of aluminium and magnesium oxides added to improve its activity. More recently, ruthenium has become the catalyst of choice, and this has helped reduce the energy requirement.

In fact, the Haber process achieves a conversion of H_2 and N_2 into NH_3 of only about 10–20% per pass through the reactor. After separation of the NH_3 product, the unconverted reactants are recycled to the reactor to obtain an overall yield of about 95%. This recycling of unconverted reactants is commonly used in industrial processes, and allows processes with low equilibrium yield to be made commercially viable.

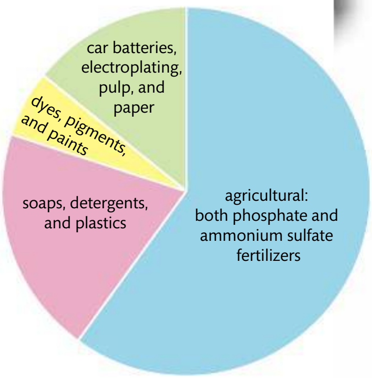

Figure 7.7 The uses of sulfuric acid.

More sulfuric acid by mass is produced worldwide than any other chemical. It has been found that the production of sulfuric acid closely mirrors historical events such as major wars that affect a country's economy. For this reason some economists use sulfuric acid production as a measure of a country's industrial strength. Sulfuric acid is used directly or indirectly in nearly all industrial processes, including the production of fertilizers, detergents, and paints and in ore processing, steel production, and water treatment. Approximately 250 million tonnes are produced annually across all continents.

The process gets its name the *Contact process* from the fact that molecules of the gases O_2 and SO_2 react in *contact* with the surface of the solid catalyst V_2O_5.

The Contact process: the production of sulfuric acid H_2SO_4

The Contact process involves a series of three simple reactions:

(i) the combustion of sulfur to form sulfur dioxide;

(ii) the oxidation of sulfur dioxide to sulfur trioxide:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H = -196 \text{ kJ mol}^{-1}$$

(iii) The combination of sulfur trioxide with water to produce sulfuric acid.

It has been shown that the overall rate of the process depends on step (ii), the oxidation of sulfur dioxide. So applying Le Chatelier's principle to this step, we can predict the conditions that will most favour the formation of product. These are summarized in the table below.

	Influence on reaction	Condition used
pressure	forward reaction involves reduction in the number of molecules of gas from three molecules reactant to two molecules product: high pressure will favour product	2×10^5 Pa (this gives a very high equilibrium yield, so still higher pressure is not needed)
temperature	forward reaction is exothermic: low temperature will increase the equilibrium yield, but decrease the rate	450 °C
catalyst	increases the rate of reaction	vanadium(V) oxide